Transferring and Reading Skills in Language Model Contexts

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ABSTRACT

When a language model benefits from a procedure supplied in context, can its internal representation replace the source document, and when does the model cease to depend on that source? We study this question by pairing two interventions: transferring states from a correct-skill run into a matched wrong-skill run, and blocking subsequent attention to the skill in the original correct-skill run. Their depth profiles reveal an intermediate regime in which skill states have lost most of their transfer effect but attention to those states remains important for successful answers. On Qwen3-8B with clinical calculators, the two half-effect boundaries occur at layers 13 and 25 of 36. The ordering holds across three models and two task domains. To establish what is being transferred, we isolate content-specific state differences and test their position, question dependence and rank. Early replacement across the skill span recovers 0.89 of the gain on a screened 135-item subset, whereas a final-position patch is strongly answer-format dependent. Recovery requires appropriate content, placement and multiple leading directions. Repeated late replacement fails to close the depth gap, and an apparent spectral collapse is explained by unusually large activations. These results separate transferability from continued source dependence and provide behavioural criteria for evaluating context representations and compression.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**; *Knowledge representation and reasoning*.

KEYWORDS

large language models, skill documents, in-context learning, activation patching, representation rank, mechanistic interpretability

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1 INTRODUCTION

Web search increasingly supplies documents directly to language models, which use retrieved content to answer questions and carry out tasks. This makes document delivery part of the computation: systems must decide what content to retain and what can be replaced by a shorter representation. Work on Web retrieval-augmented generation already shows that the representation of retrieved material affects its usefulness [38]. Procedural documents pose a particular challenge because formulas, conditions and ordered steps must support computation beyond retrieving a fact. *When can the model's internal representation stand in for a supplied procedure, and when does successful computation still require access to its source tokens?* This is the reader-side question we study, motivated by the use of external procedures in retrieval-augmented Web systems. We isolate it with controlled procedural tasks and known-correct documents, holding retrieval fixed.

We call a reusable procedure, formula, lookup table or worked example supplied in context a *skill*. Skills can substantially improve performance [22, 36], but the improvement does not reveal whether the model has formed a reusable representation or still relies on reading the document. Compact states can transfer tasks and instructions [4, 8, 17, 40], with effectiveness depending on position and task scope [23]. A failed transfer, however, has several possible causes: the probe may miss useful states, the receiving computation may be unable to use them, or the source may no longer matter. Distinguishing these possibilities is essential to interpreting what a context representation does.

Our central finding is a separation between *transferability* and *continued reading dependence*. We transplant the skill's states into a run with an unrelated, equal-length skill, then separately block attention to the skill in a correct-skill run. On Qwen3-8B and MedCalc, transfer falls below half of the available gain at layer 13 of 36, whereas blocking subsequent reading still leaves fewer than half of the rescued items correct until layer 25. There is therefore a substantial intermediate depth interval in which the states are difficult to transfer yet remain important to their original computation. This ordering holds in all six evaluated combinations of three models and two task domains. It rules out interpreting loss of transfer as the point at which the model has finished using the source.

We establish an effective transfer probe before interpreting its failure. Matched content subtraction separates the useful procedure from the effect of adding a document. Synthetic answer-format comparisons expose the limits of a final-position probe. Early span replacement then recovers 0.89 of the gain on a screened clinical-calculator subset. Content, cross-question, position and rank controls identify what supports this recovery. Repeated late replacement and spectral diagnostics constrain explanations of its loss.

For context compression and reuse, successful state transfer and safe removal of source access become separate evaluation targets. We contribute:

(1) **A paired test of transfer and reading.** Span replacement and attention knockout reveal an intermediate regime of weak transfer and strong source dependence, replicated across models and task domains (§4.7–4.8).
(2) **A controlled account of the transferred content.** Matched subtraction and answer-format tests motivate

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span interventions. Content, position and cross-question controls identify what makes early transfer effective (§4.2–4.5).

(3) **Constraints on representation compactness.** Rank truncation, repeated late injection and spectral diagnostics distinguish behaviourally useful structure from geometric summaries (§4.6–4.7).

2 RELATED WORK

Transfer interventions and source dependence. Task and function vectors transfer behaviours from demonstrations [17, 40]. Natural-language instructions can yield different function-vector components [8], and instruction-vector patches can interact superadditively across layers [4]. Transfer also depends on time and semantic scope [23]. These results motivate testing continued source dependence separately: Sia et al. [34] use cumulative context masking, and Pola and Balasubramanian [31] compare instruction-span and final-position patches. We pair span transfer with cumulative attention knockout under matched task and model conditions.

What a transferred state measures. A multiple-choice answer symbol [44] and states from completed reasoning [27] differ from the procedural context supplied before generation. Sequential patching [10] and repeated activation steering [25, 30, 35] also intervene beyond our one-time prefill protocol. Following activation-patching methodology [16, 46], we use matched controls, generated answers and identity checks to delimit what recovery establishes. Our depth comparison extends beyond tracing factual recall [13, 28] to the sustained use of procedures.

Skills and compact context representations. Skill benchmarks evaluate retrieval and application [14, 22, 36]. We hold document selection fixed and study the reader. Learned compression uses aggregated activations [2], summary tokens [29], compact states [12] or weight updates [5, 6]. We intervene on naturally occurring states without training a compressor. Dong et al. [9] study single-vector limitations in linear attention and test multiple task vectors. Their task-mapping rank differs from our rank of state differences across procedural positions.

Rank and spectral diagnostics. Spectral studies describe hidden-state geometry [11, 41] and perturbation sensitivity [32]. We instead truncate the injected matrix and measure recovered behaviour. Unusually large activations [33, 37] and dominant dimensions [21, 39] motivate a separate check of whether an apparent spectral collapse actually reflects useful-content compression.

3 SEPARATING TRANSFER FROM READING

Two intervention questions. State transfer tests recovery in a wrong-skill receiver. Attention knockout tests source dependence in the correct-skill run. Both score generated answers while varying intervention depth.

Isolating content before transferring it. Let x be a question, s its correct skill, and $\tilde{s}$ an unrelated skill with the same token length. For a fixed autoregressive model of hidden width D, let $h^L_{\text{skill}}(p)$, $h^L_{\text{wrong}}(p)$ and $h^L_{\text{no}}(p)$ denote residual-stream hidden states at layer L and prompt position p with s, with $\tilde{s}$ and without a skill. At the final prompt position, we decompose the total displacement as

$$\underbrace{h_{\text{skill}} - h_{\text{no}}}_{t\ \text{(total)}} = \underbrace{h_{\text{wrong}} - h_{\text{no}}}_{g\ \text{(presence)}} + \underbrace{h_{\text{skill}} - h_{\text{wrong}}}_{d\ \text{(content)}}. \tag{1}$$

We omit layer and position indices in Eq. 1 for readability. "Presence" denotes the change induced by the chosen wrong skill. "Content" denotes the additional change induced by replacing it with the correct skill. This algebraic decomposition does not assume independent underlying factors. We test each component by adding it to the corresponding state in a no-skill run.

Across the skill's m token positions S, the content-specific difference is a matrix $d^L_S = H^L_{\text{skill}}(S) - H^L_{\text{wrong}}(S) \in \mathbb{R}^{m\times D}$. We use a run containing the wrong skill as the *receiver*, and the correct-skill run as the *donor*. Injection at dose α sets

$$H^L_{\text{patched}}(S) = H^L_{\text{wrong}}(S) + \alpha d^L_S. \tag{2}$$

At $\alpha = 1$, the donor's states replace the receiver's across the span. Other states remain unchanged at that layer. Computation then continues. Matching lengths preserves positions and their encodings. Span interventions change many more states than final-position interventions, which limits direct comparisons. Here the final position is the prompt's end, after the query. State patches are applied during prefill, after which decoding proceeds without further patches.

Behavioural recovery and evaluation subsets. We score generated answers and measure the fraction of the donor's accuracy gain recovered by an intervention:

$$\rho(\text{arm}) = \frac{\text{acc(arm)} - \text{acc(receiver)}}{\text{acc(donor)} - \text{acc(receiver)}}. \tag{3}$$

Here, an arm is an intervention condition, and the donor baseline uses the correct skill in context. Thus $\rho = 1$ recovers the donor's observed gain, and values above one indicate greater accuracy than the donor. We distinguish rescued items (R: wrong without the skill, correct with it), persistent failures (F: wrong under both), kept items (K: correct under both), and broken items (B: correct without, wrong with). Main causal comparisons use rescued items. For span interventions, we further retain only skill groups for which the wrong-skill receiver solves none of the sampled items. We apply this screen separately within each run, model and task, without using intervention outcomes. The estimates are conditional on baseline failure in the sampled group, rather than recovery over the full benchmark.

Measuring continued reading. Cumulative attention knockout masks the skill positions as attention keys at every layer from L onward, during both prefill and decoding. On the knockout run's rescued items within the screened groups, let $q(L)$ be the fraction still solved after this intervention. A low $q(L)$ means that removing subsequent access is damaging. The transfer boundary is the first layer with $\rho(L) < 0.5$ after early successful transfer. The reading boundary is the first with $q(L) \geq 0.5$. The interval between them has weak transfer yet strong reading dependence under these thresholds. These are summaries of two curves with different intervention backgrounds, not a common measure of information content.

Testing the structure of transferable states. To test how many directions support recovery, we subtract the mean across positions, truncate the resulting matrix, and restore the mean:

$$d^{(k)} = \mathbf{1}\bar{d} + \sum_{i=1}^{k} \sigma_i u_i v_i^\top, \qquad d - \mathbf{1}\bar{d} = \sum_{i} \sigma_i u_i v_i^\top. \tag{4}$$

Here, $\bar{d} \in \mathbb{R}^{1\times D}$ is the mean state difference, repeated at every position. The singular values σ_i are ordered from largest to smallest, with u_i and v_i their associated directions. Thus k counts retained directions after centring, and the complete injected matrix has rank at most k+1. A bottom-k control keeps the k smallest singular directions and the same mean. We also vary the intervention layer. Span replacement tests whether states at that layer suffice to transfer the effect. Blocking attention to the span from that layer onward tests whether subsequent computation still depends on reading it.

4 EVIDENCE FOR TRANSFER WITHOUT SOURCE INDEPENDENCE

4.1 Experimental setup

Tasks and models. Our main evaluation pairs MedCalc-Bench [20] with skills from SRA-Bench [36], covering 55 clinical calculators with 20 questions each (1,100 items). Qwen3-8B [45] generates step-by-step reasoning using greedy decoding, without tools. We evaluate its answers with the benchmark scorer. Accuracy is 74.2% with the correct skill, 34.5% without a skill, and 31.5% with a wrong skill. For the decomposition and answer-format experiments, we use a synthetic unit-conversion task with a controlled lookup table and matched wrong skills. We also test Qwen3-0.6B, Qwen3-1.7B and Mistral-7B-Instruct-v0.3 [19], and evaluate transfer to SRA-Bench's TheoremQA subset (747 questions and 320 skills) using its prompt and scorer.

Sampling and uncertainty. The main MedCalc span experiments cover 467 rescued items from 48 calculators with at least two such items. Screening for groups whose receiver solves no sampled item retains 135 items from 14 calculators. The main depth sweep, rank curve and span-quarter comparisons use this same subset. TheoremQA uses the same screen, becoming item-level when a skill has only one question. Intervals resample skill groups, with donor and receiver baselines matched to each intervention's available items. Only complete reruns enter curves. Appendix B records earlier development samples.

Identity control. Writing the receiver's own states back should leave its output unchanged. This identity control exposed a mismatch: capture omitted an attention mask that decoding supplied, changing 108/1,766 Qwen3-8B outcomes (6.1%). Matching the masks restores exact agreement in every corrected run, for example on all 467 MedCalc items and on all 9,308 item–layer pairs of the synthetic decomposition. All main-text interventions use the correction. Appendix A.2 records provenance and Appendix A.1 implementation details. Small numerical discrepancies can propagate through long generations [15].

Table 1: Final-position state replacement across answer formats, Qwen3-8B. All formats use the same 358 items and the same skill, with every layer (0–35) tested. Recovery is the best state-replacement result over layers on each format's rescued items. The identity intervention reproduces the unpatched output on every item and layer.

Answer format	Tokens	Evaluated	Rescued	Best recovery
Multiple choice	1	358	94	1.000
Numeric answer	1–5	358	119	0.235
Chain of thought	Hundreds	358	354	0.003

4.2 Establishing a content-sensitive transfer probe

Content gain and state displacement need not agree. Across the 1,100 clinical-calculator instances, the correct skill adds 39.7 accuracy points over no skill, while the matched wrong skill changes accuracy by −2.9 points (95% interval $[-8.6, +2.9]$, paired calculator bootstrap). Yet on the synthetic task, correct and wrong skills have similar final-position displacements relative to no skill: cosine similarity is 0.91–0.95 over layers 0–10. A large shared displacement is therefore not sufficient evidence that the probe captures useful content. We test the two components separately rather than treating the total displacement as a skill.

Content subtraction recovers the behavioural effect. On the synthetic task with Qwen3-1.7B (358 items, 82 rescued), we inject each component of Eq. 1 at the final prompt position. On rescued items, the presence component recovers 0.098 of the gain, exactly the wrong-skill baseline. The content component recovers 0.805, compared with 1.000 for full state replacement. Recovery also depends on the subtraction baseline. Using a skill with corrupted numbers or with shuffled sentences gives 0.17–0.28 over layers 21–27, every layer measured, compared with 0.805 for an unrelated skill. The content component must therefore be interpreted relative to the specified wrong skill.

Single-position recovery depends on answer format. Table 1 compares answer formats on the synthetic task. All three formats use the same 358 items and all 36 layers. For short numeric answers, no final-position intervention recovers more than one of the 119 rescued items through layer 22. The best results across layers are 0.235 for state replacement, 0.21 for content and 0.008 for presence. These answers are one to five tokens long (341 of 358 use two to four). Patching the last 1, 4, 16, 32, 48 and 61 positions (61 covers the whole no-document prompt) recovers 0.235, 0.261, 0.403, 0.403, 0.395 and 0.370: the ladder saturates at 16, and giving the patch every position the prompt has does not raise it. For step-by-step reasoning, the skill raises accuracy from 0.000 to 0.989, yet single-position replacement recovers at most one of 354 rescued items (0.003) at any layer. Recovery varies with answer format, which changes both output length and computation. Same-query replacement of the final-layer final state also copies the next-token decision. We therefore interpret the comparison as a test of this one-time intervention, alongside prior evidence that transfer depends on position and task scope [23].

The corrected clinical comparison tests the final position and larger sets of positions after the skill at layer 8, using the same 135 items and wrong-skill receiver as span injection (Figure 1). The last 1, 16 and up to 256 positions recover 0.03, 0.06 and 0.05, respectively. Earlier multi-layer probes are reported separately in Appendix E.1. The complete matched comparison here concerns layer 8.

Readout checks. Generated-answer accuracy is the primary readout. Gold-answer log-probability can favour an intervention with worse accuracy, and forcing an answer cue can suppress the skill's benefit. Multiple-choice controls also retain substantial performance after content corruption [3] (Appendix D).

4.3 Recovering the effect at its source positions

Changing the intervention positions restores recovery. We next apply Eq. 2 across the skill's own positions. At layer 8, span injection raises accuracy from 0.000 to 0.815 on the screened 135-item subset, recovering 0.89 of the donor's gain (Table 2). Figure 1 compares this with writing donor states at the final position, the final 16 positions, or up to 256 positions after the skill, using the same layer, items and receiver. All positions outside the chosen intervention retain their receiver states.

Dose, transfer, order and noise controls. Recovery changes nonlinearly with injection strength. On the screened subset ($n = 135$, 14 calculators), $\alpha = 0.4, 0.5, 0.55$ gives $0.04, 0.04, 0.14$ recovery, while $0.6, 0.7, 1.0$ gives $0.53, 0.87, 0.89$. Doubling the injection to $\alpha = 2$ reduces recovery to 0.77. Most of the increase occurs between 0.55 and 0.7. To test content specificity, we compare correct and other-skill injections over a common prefix length: the correct prefix recovers 0.65, whereas content from an unrelated skill recovers 0.05. A same-family mean control recovers 0.12 on its 41 eligible items. Permuting positions also gives 0.05, norm-matched Gaussian noise gives 0.02, and the identity control gives 0.00. These controls show that recovery depends on the injected content and its arrangement across positions, rather than perturbation size alone.

These results are conditional on the receiver screen. On excluded groups, other-skill and noise controls perform better, limiting extrapolation to the full benchmark. Span patches also write many more states than a final-position probe. We address this size difference next.

4.4 Distinguishing position budget from placement

Span injection writes 365–1,630 positions, whereas the final-position probe writes one. We compare position selections at fixed written-token budgets B, using the same 135 items, layer and receiver as Table 2 (Figure 2). *Quantity alone is insufficient.* Evenly spaced or random subsets recover at most 0.15, even at half the span. Choosing the positions with the largest content change recovers at most 0.12. Writing after the span stays at or below 0.07, at layer 8 or the seven tested layers from 4 to 30. *Block selections differ.* Selecting 256 positions as one block recovers 0.37. Selecting them as 2, 4, 8, 16 and 32 spaced blocks gives 0.24, 0.22, 0.13, 0.08 and 0.08. The one-versus-32 difference is 0.29 $[0.09, 0.48]$, with calculator-bootstrap uncertainty. These selections have similar mean energy shares (0.291 versus 0.295), but energy is not matched per item and the selected content changes: procedure coverage is 38.6% versus 30.7%. This does not isolate contiguity from semantic coverage. *Section and placement matter.* A half-span block centred on the procedure recovers 0.82 $[0.67, 0.93]$, versus 0.18 around the example. Shifting each content difference by one position gives 0.15–0.22 with or without wrap-around. Omitting the first or last 16 positions but leaving the rest in place preserves 0.89–0.92. Recovery is thus sensitive to which content is injected and where, beyond the number of written positions.

4.5 Testing whether transfer carries reusable content

A successful same-question patch could carry a task-specific answer rather than reusable procedural content. Table 3 tests this alternative by varying the injected section and the donor question. Of four contiguous span quarters, the formula quarter gives the largest standalone recovery (0.56), compared with 0.89 for the full span. With the skill before the question, causal masking prevents its states from depending on later question tokens: states are identical across instances at equal prompt length and nearly identical otherwise (cosine ≥ 0.994 over all 48 calculators). Placing the skill after the question allows such dependence. Even then, states from other instances of the same skill recover 0.98 at layer 8, close to the current instance's 1.02. At layer 16, the gap widens to 0.29 versus 0.49, while the instance-specific remainder recovers little on its own. Early transfer therefore relies largely on content shared across questions, with greater question dependence later.

Two completed corrected controls test whether a shared shift or a change in state norm could explain recovery. Averaging content differences across skills recovers only 0.03 with the skill first ($n = 135$) and 0.04 with it last ($n = 151$). Rescaling patched states to match the receiver's norm at each position preserves 0.87 recovery, compared with 0.89 without rescaling. In the corrected rerun, layer-16 recovery is 0.49 with the skill last (151 items, 17 calculators) and 0.06 with it first (135 items, 14 calculators). Because these screens retain different subsets, the contrast suggests sensitivity to prompt order but is not a matched estimate of its effect (Appendix F.4).

4.6 Testing a compact representation of the span

We next inject the truncated matrix in Eq. 4 to measure how recovery depends on the retained directions. Each intervention preserves the same skill-specific mean at every span position, so the comparison tests how much variation across positions is needed beyond that mean. At layer 8 on the screened MedCalc subset (135 items, 14 calculators), recovery is $0.07, 0.05, 0.12$ for $k = 4, 16, 32$, rising to 0.63 at 64 and 0.77 at 128, and reaching the untruncated 0.89 at 256 (Figure 3). Keeping the bottom 4 to 128 directions recovers at most 0.04. Recovery therefore depends on which directions are retained: the leading directions support most of the measured effect, but a small number is insufficient under this intervention.

For comparison across models, we summarise each curve by k^*: the first rank reaching halfway from recovery at $k = 16$ to the

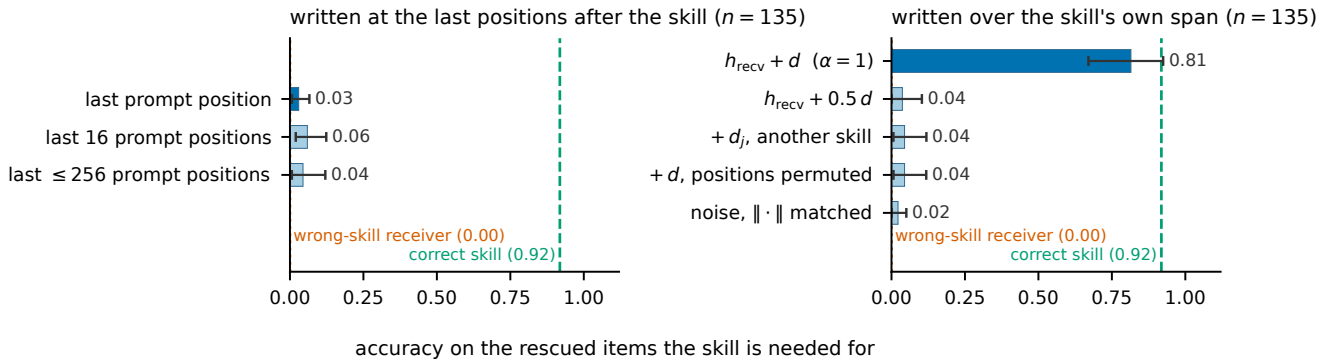


Figure 1: Matched position comparison on MedCalc with Qwen3-8B after the fix. Both panels use layer 8, the same wrong-skill receiver and 135 items from 14 calculators. Left: donor states written at the last 1, 16, or up to 256 available positions after the skill. Right: interventions over the skill span. The dashed line marks correct-skill accuracy; the dotted line marks receiver accuracy. Error bars are calculator-bootstrap 95% intervals.

Table 2: Span interventions on the screened MedCalc subset: 135 items from 14 calculators, selected from 467 rescued items across 48 calculators by requiring the wrong-skill receiver to solve no sampled item in a group. The same-family control is available on 41 items. Each row uses baselines matched to its own items in Eq. 3; accuracy intervals resample calculators. Table 4 gives the interval for full-span recovery.

intervention	n	accuracy	CI_{95}	recovery ρ
correct skill in context	135	0.919	[0.844, 0.976]	+1.00
$h_{recv} + d$ (full span)	135	0.815	[0.670, 0.925]	+0.89
correct content, shared prefix	135	0.593	[0.336, 0.820]	+0.65
half dose ($\alpha = 0.5$)	135	0.037	[0.000, 0.103]	+0.04
another skill (individual donor)	135	0.044	[0.007, 0.119]	+0.05
mean from another skill family	135	0.044	[0.012, 0.102]	+0.05
mean from the same skill family	41	0.122	[0.000, 0.250]	+0.12
positions permuted	135	0.044	[0.007, 0.118]	+0.05
norm-matched Gaussian noise	135	0.022	[0.000, 0.051]	+0.02
identity intervention	135	0.000	[0.000, 0.000]	+0.00
wrong skill, unpatched	135	0.000	[0.000, 0.000]	+0.00

maximum on the fixed grid $\{16, 32, 64, 128\}$, with interpolation in $\log_2 k$. We call this the *rank knee*, a summary of the measured recovery range. Qwen3-8B gives $k^* = 47$ [43, 55], approximately 1.1% of its hidden width. Appendix G.1 reports development and deeper-layer curves. These measurements describe what the injected representation can recover, rather than directly measuring the model's unmodified computation: truncated states may activate pathways absent from an unpatched run [26].

4.7 Identifying the interval between transfer and reading

The preceding controls establish effective early transfer of content across skill positions. We now ask whether losing that effect at later layers means the model no longer needs the document. Figure 4 tests this inference directly.

Transfer weakens while reading remains important. Figure 4 compares transfer through span replacement with continued use through attention on Qwen3-8B. Every layer is measured, and both interventions are read on the same screened items. Replacement recovers 0.92–0.98 through layer 7, 0.89 at 8 and 0.74 at 12 of 36, then 0.16 at 13 and at most 0.07 from 14 onward ($n = 135$, 14 calculators). The drop between layers 12 and 13 accounts for most of the decline. We define the crossing below $\rho = 0.5$ as the *transfer boundary* (labelled "handover" in the figures and tables). The log-probability of the donor's generated reasoning also drops between these layers, suggesting that binary answer grading sharpens, but does not fully explain, the transition (Appendix B). To test continued reading, *cumulative attention knockout* blocks attention to skill tokens at every layer from L onward. On the 134 rescued items of the same subset, 0.03–0.12 stay correct for $L \leq 23$. This rises to 0.25 at 24, first passes half at 25 (0.58) and reaches 0.94 at 35. The transfer boundary therefore lies at $13/36 = 0.36$ of depth and the reading boundary at $25/36 = 0.69$, using the same screen and restricting knockout to its donor-correct items. Writing the last prompt position instead recovers at most 0.06 at any layer. Continued dependence on skill attention thus outlasts successful transfer from the skill's states. Transfer measures recovery in a

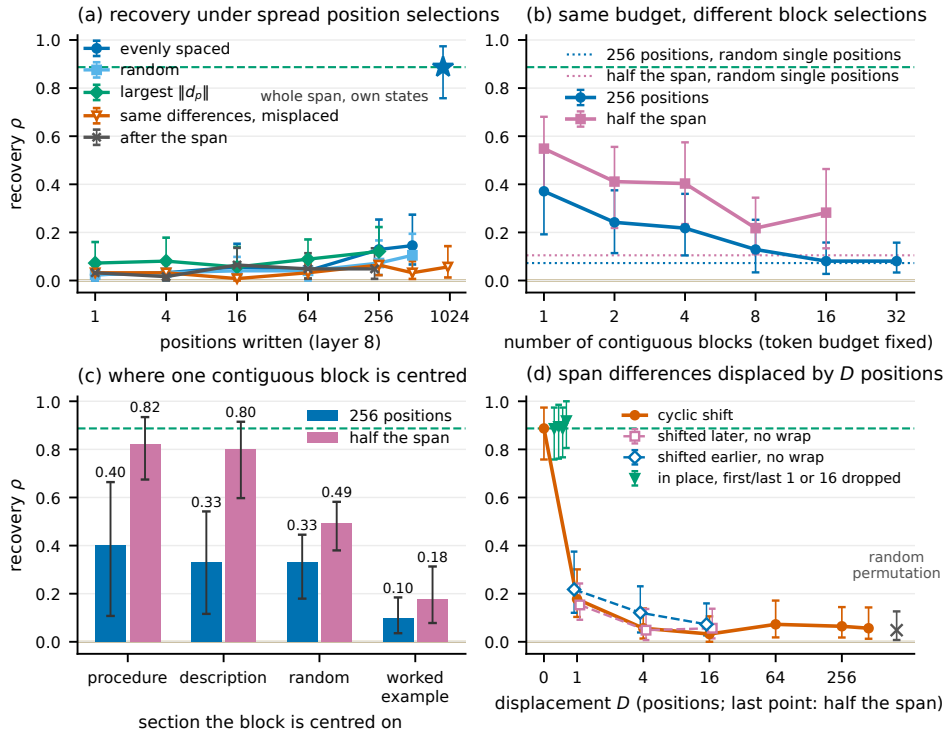


Figure 2: Position budget on MedCalc with Qwen3-8B, layer 8: the same 135 screened items (14 calculators) and wrong-skill receiver in every panel; the dashed line is the whole span written in place ($\rho = 0.89$). (a) B positions, each written with its own donor state (evenly spaced, random, or largest $\|d_p\|$), the corresponding differences written half a span away, or the last B prompt positions after the span. (b) A fixed budget written as N contiguous blocks; dotted lines are random single positions at the same budget. (c) One contiguous block centred on a section of the skill. (d) The span differences displaced by D positions, cyclically or without wrap-around; triangles write it in place without its first or last 1 or 16 positions. Error bars are calculator-bootstrap 95% intervals.

wrong-skill receiver, whereas knockout measures dependence in a correct-skill run. The separation identifies the intermediate regime operationally: from layers 13 through 24, span replacement recovers less than half the gain and blocking further reading leaves fewer than half the rescued items correct. From layer 25, at least half remain correct without further reading, with residual dependence continuing to diminish. Thus transfer loss precedes, rather than demonstrates, tolerance to removal of source access.

Repeated replacement does not bridge the gap. To test whether late failure reflects intervening at too few layers, we replace span states repeatedly within consecutive layer windows, using the battery's receiver and the same 135 items as the depth sweep. Windows 8–11, 12–15, 14–19 and 16–35 recover 0.89, 0.74, 0.16 and 0.06. The last equals layer 16 alone. Writing every layer at once recovers 0.98 [0.91, 1.03], so the transplant itself remains effective when it starts early enough. The identity control is exact on all 467 items. On TheoremQA (99 items, 75 groups, with its battery's receiver), the same windows recover 0.84, 0.61, 0.38, 0.20, the last again equal to layer 16 alone. Repeated late injection therefore does not restore early-layer recovery in these tests. This constrains a simple shortage-of-interventions explanation, while leaving interactions with the receiver's other states open [4].

A spectral drop is not evidence of compression. Could the loss of transfer reflect compression into fewer dimensions? We examine the centred participation ratio, a summary of how broadly matrix energy is distributed across singular directions. It falls from 68 at layer 12 to 1.0 at layer 16, but this drop is driven by unusually large activations [33, 37]. In 35 of 48 calculators, one token's state reaches 121–195 times the median span norm, always on the same two hidden dimensions. Removing the four highest-norm positions restores the ratio to 85 at layer 16, and it stays between 62 and 98 over layers 0–28. The drop is also absent in Qwen3-0.6B and Qwen3-1.7B. This diagnostic therefore does not support compression as the explanation for lost transfer: a few large activations distort the dimensionality summary (Appendix Table 13).

4.8 Replication across models and task domains

We extend span injection, rank truncation and the available depth comparisons to Qwen3-0.6B ($D = 1024$), Mistral-7B ($D = 4096$,

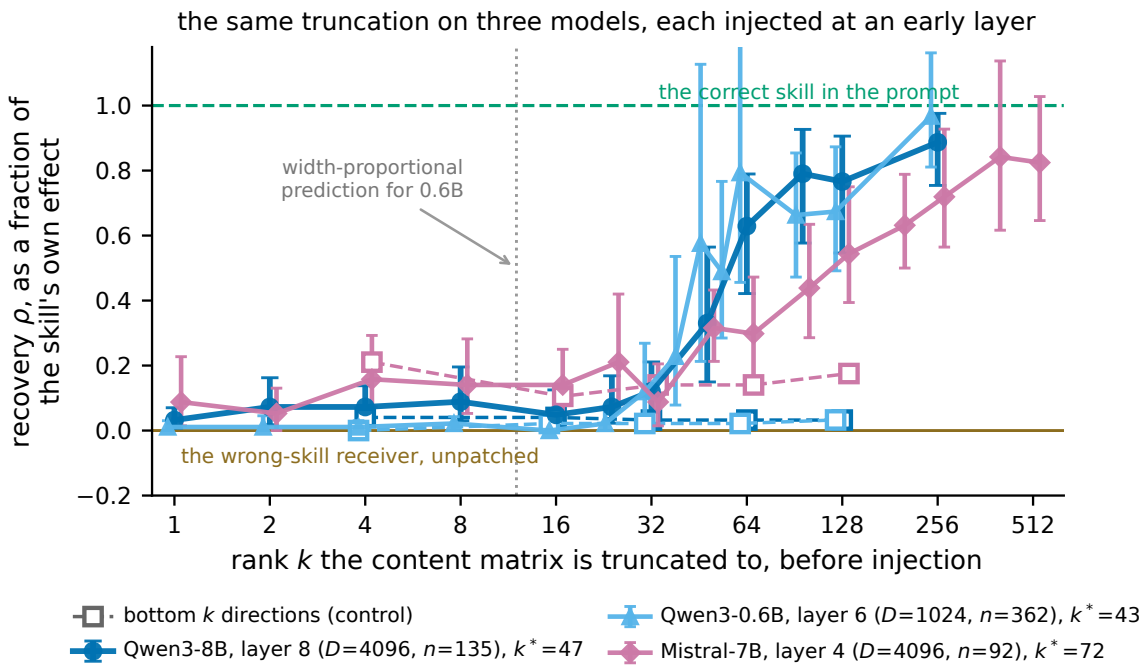


Figure 3: Recovery after retaining k leading centred directions and restoring the positional mean, on MedCalc after the fix. Each model is injected at an early layer. Despite a fourfold difference in hidden width, Qwen3-0.6B and Qwen3-8B have similar rank knees (43 and 47); Mistral-7B, with the same width as Qwen3-8B, has $k^* = 72$ [46, 94]. Untruncated recovery is 0.89, 0.91 and 0.84, respectively, on each model's screened subset (135, 362 and 92 items). Both Qwen models reach that value by $k = 256$ and Mistral-7B at $k = 384$. Dashed curves retain bottom directions. Error bars are calculator-bootstrap 95% intervals. Only complete rank experiments are plotted.

Table 3: Content and question dependence of span states, Qwen3-8B after the fix. Top: each contiguous quarter is injected separately, on 135 items from 14 calculators. Bottom: with the skill after the question, content shared across other instances and the instance-specific remainder are injected separately, on 151 items from 17 calculators. Each comparison uses its screened subset from the larger rerun. All entries report recovery ρ.

Skill quarters, layer 8, skill before question ($n = 135$)	
Injected content	Recovery ρ
Prose description	0.09
Formulas	0.56
Tool signatures	0.12
Worked example	0.05
Full span	0.89

Question dependence, skill after question ($n = 151$)		
Injected content	Layer 8	Layer 16
This instance	1.02	0.49
Other instances of the same skill	0.98	0.29
Instance-specific remainder	0.03	0.05

32 layers), and TheoremQA. Each model is injected at an early layer, and the receiver screen is applied separately to each run. Table 4 collects the paired transfer and reading boundaries alongside the early-transfer and rank results. Results for Qwen3-1.7B, where skills rescue only 8–11% of items, appear in Appendix Table 14.

Span recovery ranges from 0.74 to 0.94 across the six model–task combinations. The largest control is 0.35, for a same-family mean on 31 Mistral/MedCalc items, where full-span recovery on the same items is 0.71. In every model–task combination, the transfer boundary occurs before the reading boundary. The measured transfer boundaries lie at 0.33–0.43 of model depth within Qwen and at 0.19–0.22 in Mistral. Reading boundaries also vary by task: 0.58–0.71 on TheoremQA and 0.69–0.84 on MedCalc across the three models.

The pre-registered width-proportional prediction places Qwen3-0.6B's rank knee near 12. Instead, it is 43 [39, 52], close to Qwen3-8B's 47 [43, 55] despite the fourfold width difference. TheoremQA gives 52 [46, 62] and 50 [39, 67], respectively. Mistral gives $k^* = 72$ [46, 94] on MedCalc and 64 [50, 79] on TheoremQA. These values are higher than Qwen3-8B's but have overlapping intervals. Thus hidden width alone does not determine the measured rank requirement, while a difference between families is not established. Full curves and controls are retained in Figure 3 and Appendix G.1.

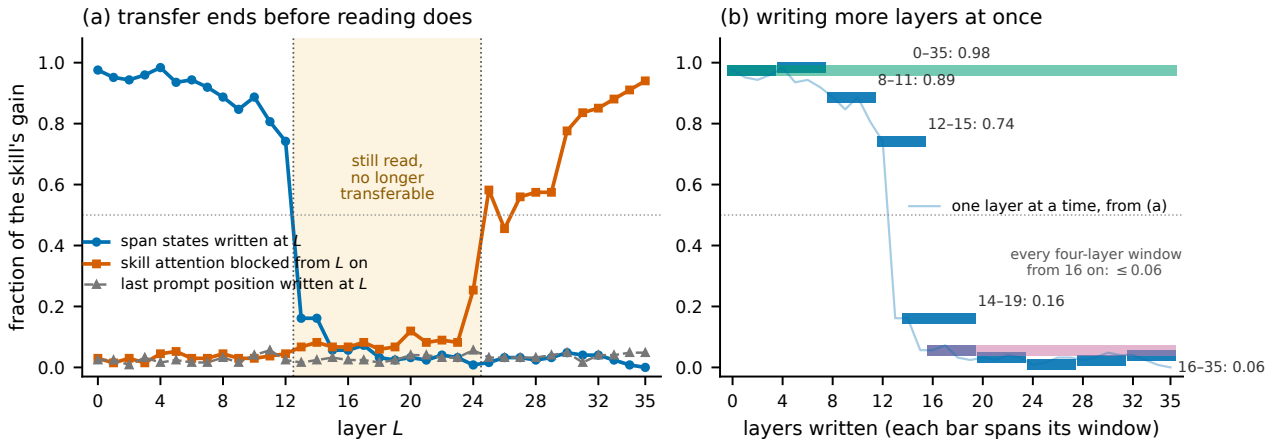


Figure 4: Transfer declines earlier than dependence on skill attention, Qwen3-8B on MedCalc, with every layer measured. (a) Three interventions on one axis and the same screened items (n = 135; the knockout on their 134 rescued items): span states written at layer L; attention to the skill blocked from L onward; and the last prompt position written at L. The shaded band spans the layers where skill attention still supports performance but span replacement recovers less than half of the gain. (b) Each window is drawn across the layers it writes: four-layer windows tile the stack, and the long windows 14–19, 16–35 and 0–35 are shown with them; the faint curve repeats (a). Writing every layer at once recovers 0.98, while any window starting at 16 or later stays at or below 0.06. Span interventions use the corrected implementation; attention knockout does not use state capture.

Table 4: Replication with the corrected implementation. Injection layers are 8 for Qwen3-8B, 6 for Qwen3-0.6B and 4 for Mistral-7B. Recovery intervals resample skill groups. Max control includes half-dose, other-skill, permutation and noise comparisons, including same-family means available on 41 MedCalc/Qwen3-8B and 31 MedCalc/Mistral items. n describes the battery. Handover brackets the transfer boundary as a fraction of model depth; reading is the first layer, at one-layer resolution, from which blocking attention to the skill still leaves at least half of the rescued items correct, measured on the knockout run's rescued items within the same screened groups. k^* is the rank knee on the fixed grid.

task	model	n (groups)	ρ [95% CI]	max control	handover	reading	k^*
MedCalc	Qwen3-8B	135 (14)	0.89 [0.76, 0.97]	0.12	0.33–0.36	0.69	47
TheoremQA	Qwen3-8B	99 (75)	0.84 [0.75, 0.92]	0.25	0.33–0.36	0.58	50
MedCalc	Qwen3-0.6B	362 (38)	0.91 [0.75, 1.05]	0.30	0.36–0.39	0.79	43
TheoremQA	Qwen3-0.6B	132 (97)	0.74 [0.66, 0.82]	0.13	0.36–0.39	0.71	52
MedCalc	Mistral-7B	92 (20)	0.84 [0.69, 1.09]	0.35	0.19–0.22	0.84	72
TheoremQA	Mistral-7B	115 (94)	0.94 [0.84, 1.05]	0.27	0.19–0.22	0.59	64

5 DISCUSSION AND CONCLUSION

What the depth gap establishes. The paired tests distinguish early effective transfer, an intermediate interval of weak transfer with continued reading dependence, and later tolerance to blocking further reading. These operational regimes do not identify separate internal algorithms. The gap is compatible with dependence on the receiving state, whose role remains unresolved.

Implications for representation research. What a state can transfer and whether its source remains necessary are different questions. Content and cross-question controls identify useful procedural information, while position and rank interventions characterise its usable structure. Failed late transfer and distorted spectral summaries cannot alone establish that this information has disappeared or been compressed. Pairing transfer with source-access interventions provides a behavioural way to distinguish these claims.

Implications for context compression and reuse. Representations proposed for reuse should be tested both for recovery in the intended receiver and for tolerance to removing source access. Our findings show why token count, state norm and single-format success are insufficient proxies, and why hidden width alone need not determine a suitable rank budget. These are evaluation principles, not demonstrated efficiency gains. The evidence concerns fixed contexts and screened subsets. Layer depth does not specify agent steps or establish when documents can be dropped during multi-turn execution.

AI USE STATEMENT

Generative AI tools assisted with manuscript restructuring and language editing, literature lookup and reference checks, consistency checks between reported results and local experiment records, and

figure-label editing. The authors are responsible for the final text, scientific claims and experimental artifacts.

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A SETUP, IN FULL

Two task families. Real skills, the main material: MedCalc-Bench instances with the 55 gold calculator skills released with SRA-Bench—55 calculators × 20 instances = 1,100 items, graded deterministically against a numeric tolerance with no model in the loop. Both the skills and the tasks are other people's, which is what makes the effect size credible. Qwen3-8B, single turn, no tools, greedy. Behavioural baseline and replication intervals resample skill groups because items sharing a skill are dependent; original item-level intervals in individual tables and plots are identified separately in the main text. *A synthetic positive control*: a fictional unit system with a one-page document giving its conversion table, so the no-document floor is guaranteed, contamination is impossible, and we know exactly what content means. Items are generated from the table ($n = 358$), and every item whose answer appears in a worked example is dropped.

Two dependent variables. Accuracy, from a greedily decoded free-form answer graded by the task's own scorer, and the teacher-forced log-probability of the gold answer. They fail differently and disagree often enough that reporting one alone is unsafe (§4.2); accuracy is read first throughout. No item was rewritten into multiple choice to obtain an accuracy channel—§ C.1 shows why that rewrite is not harmless.

Interpreting recovery. Equation 3 normalises an intervention by donor and receiver accuracy on the same items. It measures the fraction of the available gain, not raw accuracy or a percentage of information transferred. Values can exceed one. Comparisons across protocols must retain their own paired baselines.

The instrument's own noise floor, measured. Every decoding run carries a *null* arm: the receiver's own span states are captured and written back over themselves. With a length-matched receiver this is a bit-for-bit identity, so the null arm must equal the unpatched receiver at every layer. Before the fix of Appendix A.1 it did not, quite. On Qwen3-8B, across 1,766 paired observations from every pre-fix run that carries both arms, the null arm agrees with the receiver on 93.9% of items: 108 flip, 46 from wrong to right and 62 from right to wrong, symmetric within a sign test ($p = 0.15$). The pipeline itself is deterministic—the same configuration run on two different machines agreed on all 360 item-by-arm cells—so this is not run-to-run variation but a reproducible property of the instrument: the capture and decoding passes took different attention kernels, which perturbs the written-back state by 1.2% at layer 8, and the dependent variable is the end of a chain of thought several hundred tokens long, which is long enough for that to change an argmax somewhere and propagate. It scales with the model: the same arm flips 0.6% of items on Qwen3-0.6B (1,696 pairs) and 0.8% on Qwen3-1.7B (1,338), against 6.1% on the 8B. After the fix it is exact (80/80).

Three consequences are used throughout. First, **an effect smaller than about six points is not resolvable on the 8B** in pre-fix runs, whatever a binomial interval says. Second, the floor is not uniform: on the persistent-failure cell it is 16%, between two and three times the average, which is what one expects of items on a decision boundary, so numbers read on that cell are read against a coarser ruler. Third, on the restricted split the pre-fix null arm read 0.034 rather than zero—the split conditions on the receiver failing, so a flip there can only go upward—so pre-fix span numbers are read against 0.034; after the fix it reads exactly 0.000, the floor Table 2's 0.750 is read against. We have rarely seen this control reported in the patching literature; at a single answer token it is small enough to be invisible, and it becomes large only once the dependent variable is a long generation.

Four cells. Every item is labelled by what the document did to it: **R** (rescued: wrong without, right with), **F** (persistent: wrong both ways), **K** (kept: right both ways), **B** (broken: right without, wrong with). **R is the only cell in which the document demonstrably works**, every table here is read by cell, and §4.2 shows the aggregate over all four can be sign-inverted relative to R.

A.1 Experimental detail

Models and numerics. Qwen3-1.7B (28 layers, $d = 2048$) and Qwen3-8B (36 layers, $d = 4096$); §4.8 adds Qwen3-0.6B (28 layers, $d = 1024$) and Mistral-7B-Instruct-v0.3 (32 layers, $d = 4096$), whose chat template has no system role, so the system text is merged into the user turn. Greedy decoding throughout, `enable_thinking` disabled and verified by a self-test on the rendered template. Eager attention wherever an attention mask is manipulated, because sdpa and flash silently ignore a 4D mask; layer-output hooks are implementation-independent, so the decode-heavy runs use sdpa, which is about 2.5× faster on a 2k-token prompt. The 1.7B synthetic runs are fp32: bf16 costs 0.5–2.5 nats on the log-probability of a token deep in the tail, and every log-probability quantity here is a difference of such numbers. The 8B runs are bf16 and the gaps we report from them are 5–8 nats, well outside that error; the accuracy channel is immune to it either way.

Self-tests, and one that failed. Three checks run with the experiments. (i) *Self-transplant*: donor and receiver are the same forward, so the patch must be an exact no-op. On the fp32 synthetic runs the deviation is 0.0 nats at every layer. On the bf16 8B runs it is up to 0.38 nats—bf16 arithmetic over two sequence lengths—while the accuracy channel remains an exact no-op at every layer, which is how we tell the two apart. (ii) *Last-layer patch*: replacing the final layer's output at the patched position makes that position's logits the donor's by construction, so the next token must match exactly. (iii) *Query-side control* for the knockouts.

An error worth recording. The first version of the real-skill scoring code patched at index −1 of a forward run over prompt *and* answer, which is the last token of the answer—after every position whose logits are scored. Every arm returned the untouched baseline to six decimals, which is what made it visible. Relative indices are the wrong default whenever the forward length varies between the capture and the patch.

Control documents. The synthetic controls are generated from the skill document by `make_controls.py`, which checks two things: the corrupted document's numbers must not coincide with the task set's gold answers (7 of 249 do, 2.8%), and the shuffle must not be the identity. All three controls are stored under the same filename in separate directories, because the injector derives the "`# Skill: name`" header from the file stem—a file called `SKILL.zorb-units.corrupted.md` would announce the manipulation in the prompt, and the control would then differ from the treatment in two ways.

Item set. The synthetic set is generated from the conversion table with the value range 2–40, dropping any item whose answer also appears in a worked example of the document ($n = 358$; the sweeps use the first 120 or 200). Distractors are values reachable by misreading the table—the neighbouring row, the wrong family, the inverted direction—never random numbers, and the key is rotated across A/B/C/D so a position-biased model still scores at chance.

A.2 Evaluation scope and intervention provenance

An accidental replication. A queueing mistake ran one 36-layer synthetic sweep (358 items, two arms, generations up to 900 tokens) twice, on two different GPUs in two different containers. All 36 layer files agree byte for byte. This does not bound the nondeterminism discussed in §4.1—it is one configuration on one pair of devices—but it is the only end-to-end check of it we have, and it came out exact.

Sample size and scope. The battery was developed on $n = 40$ (ten calculators × four rescued instances), which gives intervals of roughly ±0.15; §4.3 reruns it on forty calculators and §4.8 on further models and a second task, and the conclusions are the ones that survived that. We report profiles and not individual layers: the one non-monotonicity in the depth curve (0.250 at layer 18 against 0.475 at 20) has overlapping intervals and we do not interpret it. Among the three models, Qwen3-1.7B is the weakest rung: the document rescues only 8–11% of its items, so its truncation curve is not resolved and is reported for completeness rather than relied on. The width argument rests on Qwen3-0.6B against Qwen3-8B, a factor of four within one family; a second family of the same width needs more directions (Appendix G.2). In the prefix development runs, four independent shards recomputed the three baselines and agreed to the item (0.950 / 0.025 / 0.175), and the no-op arm sits on the receiver's baseline at all twelve layers to within the 5% numerical floor of §4.1, so the instrument is checked even where the estimate is coarse. The span channel, the depth sweep and the truncation curve have three models and two tasks; the knockouts two models and two tasks; the final-position results two models (Qwen3-1.7B, fp32); the document-order comparison and the within-document localisation one.

Which items the causal claims are about. Two restrictions stack. Causal claims are made only on the rescued cell, the only cell in which the document demonstrably works; on the persistent-failure cell the same arms track the document rather than correctness—the transplant fails exactly where the document fails—which is the right behaviour but is not a substitute for the other cells. And within that cell the span arms are read on the calculators whose *receiver* never solves an item, because on the others the model does not need the document, and before the fix any disruption of the wrong one, noise included, let it fall back on what it already knows ($\rho = 0.14$ for noise there against 0.00 here). After the fix noise no longer does that (0.00 on those calculators), but a donor from another skill still reads 0.47 there against 0.19 on the rest. On forty calculators the restriction keeps fifteen of them and 60 items. On this split, accuracy is 0.750 for the transplant and at most 0.083 for the tested negative controls; the main text distinguishes item-level intervals from group-bootstrap uncertainty. Those forty calculators are themselves the 160 rescued items with the most instances per calculator, out of 467. We therefore reran the battery, the truncation curve, the depth sweep, the dose curve and the within-document quarters over *every* rescued item of every calculator with at least two (467 items, 48 calculators; the restriction then keeps 135 items over 14). Every conclusion holds and no control moves above 0.12; Appendix B gives the comparison arm by arm.

What the rank result does and does not establish. §4.6 truncates the injected content matrix to rank k at a layer where the untruncated version recovers most of the effect, and recovery stays near the control floor at $k \leq 32$ while the bottom-k control stays at the floor. That licenses the forward direction: the effect requires directions of the same order as the span's participation ratio, and within the Qwen family neither number scales with width. It does not license a claim that the number belongs to the document alone (Mistral-7B needs more), nor a converse about depth. Supplying a deep layer with a higher-rank message taken from a shallow one does not restore the handover ($\rho = 0.00$ against 0.06 for the same-layer donor, 0.00 with the norm matched), so we do not claim that what a deep layer lacks is rank.

Diagnostic results for the behavioural readout. We could not find any representational quantity at the final position that distinguishes the items a skill rescues from the ones it does not: the norms and angles agree to three digits, and the one similarity measure that separates them under the gold skill separates them equally under a wrong one. And the document acts through a chain of thought, so every causal measurement decodes several hundred tokens; scoring under a forced `ANSWER:` cue instead leaves almost no behavioural effect (0.178 with the skill, 0.131 without and 0.100 with a wrong one, on 720 items) while retaining a +1.08 nat log-probability difference—so the span channel has no usable log-probability reading, and is reported on accuracy alone.

An instrument fault and result provenance. State capture omitted the attention mask supplied during decoding. Although mathematically equivalent for an unpadded sequence, these calls selected different kernels. Their states differed by 1.2% at layer 8 and 25% at layer 34, and identity patching changed 6.1% of Qwen3-8B answers. Matching the masks restores exact identity agreement. All main-text state interventions use the corrected implementation, including the synthetic and final-position sweeps (§4.1). Attention knockout does not use state capture. Historical development results in the extended appendices retain their original sample sizes

and provenance. The corrected full-data rerun in Appendix B supersedes the earlier 60-item battery.

Numerics. Log-probabilities on the synthetic control are fp32, because bf16 costs 0.5–2.5 nats on a token deep in the tail. The 8B numbers are bf16, which appears as self-transplant deviations of 0.1–1.1 nats while the accuracy channel remains an exact no-op; accuracy is read first throughout.

B THE FULL-DATASET RERUN, AND WHAT READS THE SPAN

Every MedCalc × Qwen3-8B result in the main text was developed on the forty calculators with the most rescued items, sampling four each: 160 of 467 rescued items, and 60 items over 15 calculators after the receiver-never-solves restriction. We reran each of them with every rescued item of every calculator that has at least two (467 items, 48 calculators), changing nothing else. The restriction is stricter on the larger set—a calculator now has to fail on up to twenty items rather than four—so it keeps 135 items over 14 calculators, and we also report the per-item restriction (369 items over 45), which keeps the items whose own receiver fails.

The transplant, the truncation knee between 32 and 64 directions, and the formula quarter's dominance all reproduce. The one arm that moves is the same-family donor, from 0.05 to 0.12, on the 41 items where it is defined.

The depth sweep at one-layer resolution. The subset sweep measured layers 0, 4, 8, 12, 14, 15, 16, 20 and placed the collapse somewhere after 12. At full scale we added 13 and 17–19: recovery is 0.98, 0.98, 0.89, 0.74 at layers 0, 4, 8, 12, then 0.16, 0.16, 0.06, 0.06, 0.07, 0.03, 0.02, 0.03 at 13 through 20 (per-calc; the per-item restriction gives 1.00, 0.98, 0.94, 0.79 then 0.32, 0.34, 0.25, 0.25, 0.23, 0.19, 0.14, 0.17). Layers 13 and 14 are indistinguishable under both restrictions, so the transition is a single layer between 12 and 13 and not the gradual decline a stride-2 sweep suggests.

A continuous dependent variable at the same boundary. Accuracy over a chain of thought is a threshold, and a smooth decline inside the model could present as a step in it. We therefore decode the with-document trajectory once per item and teacher-force that same trajectory under every other condition, reading its mean token log-probability: $q = (\ell_L - \ell_{\text{recv}})/(\ell_{\text{gold}} - \ell_{\text{recv}})$ has the same two anchors as ρ with no decoding threshold anywhere in it ($n = 442$ items on which the document run is correct, 48 calculators). q reads 0.999, 0.989, 0.960, 0.831 at layers 0, 4, 8, 12, then 0.598 [0.550, 0.648] at 13, 0.595 at 14, 0.340 at 16, 0.201 at 20 and 0.129 at 24. Expressed per layer as a fraction of its range, the step at 12 → 13 is 26.8%, against 3.7% per layer over 8–12 and 0.3% from 13 to 14: the discontinuity is present in a readout that cannot be produced by grading. It is smaller than the behavioural step (61% of range), so the grader sharpens the boundary roughly twofold without creating it, and it is spread evenly across the four quarters of the trajectory (0.20, 0.26, 0.31, 0.21) rather than concentrated at the answer, which is the opposite of what a grading artefact predicts. We also tried restricting q to the tokens after the final answer marker and report that it carries no information here: once the full reasoning is teacher-forced, every condition predicts the final number almost perfectly (median ℓ between −0.001 and −0.005 for all of them, and the gold-minus-receiver denominator has median 0.0014, below 0.01 on 59% of items).

Which heads read the span. We recorded each head's contribution from the span to later positions, $W_O^h \sum_{s \in \text{span}} a_{t,s} v_s$, comparing the document and filler runs ($n = 455$). Raw magnitudes are dominated by the last layers, but the residual stream grows by more than an order of magnitude with depth; normalised by it, the read-out concentrates in layers 12–21 and peaks at 18. It is not a small circuit: of the 1,152 heads the top 1% carry 10.5% of the read-out and the top 5% carry 31.1%, and no head attends to the span more than uniformly (0.4–1.2×, with the span occupying half the prompt). Blocking the 32 highest-ranked heads' attention to the span—ranked on half the calculators and evaluated on the held-out half—costs 0.39 of the document's effect, against 0.23 for 32 random heads at the same layers (paired difference 0.154, 95% CI [0.061, 0.251], McNemar 39 versus 9 discordant, $n = 195$ over 23 calculators). The ranking is therefore causal, but it buys only 0.15 over chance and 32 heads remove well under half the effect: the span is read diffusely, and we do not claim a nameable set of reader heads. One further observation constrains what fails at depth. After an injection at layer L, the span's own states from $L + 1$ onward are identical to the document run by construction—the span attends only to a shared prefix and to itself—and the downstream per-head read-out of the span is indeed 0.86–0.99 of the document's at every injection layer we tried, including those at which behaviour is gone. This read-out is compatible with continued access to donor content. A late span patch leaves the receiver's earlier question-state updates unchanged, motivating a direct test of their interaction with the transplanted states.

C CONTROL DOCUMENTS IN FULL

d is defined by the document it is subtracted against, so "content" is whatever separates the skill from that control. Our default control holds only length fixed. Two tighter ones, generated from the skill document itself, hold more: **shuffled** keeps every word and every number and destroys the line order; **corrupted** keeps the wording, the procedure, the table layout and the ordering of the units, and replaces *every conversion factor*. All three are injected under the same name and header, so the prompt never announces which it is.

Both tighter controls move the accounting a long way (Table 6). A table whose numbers are all wrong solves 0.720 of the rescued cell; a table with the right numbers in scrambled order solves 0.659; the unrelated text solves 0.098. Against no document the cell is 0.000 by construction and with the gold document it is 1.000. So when either tight control is used, g is no longer inert—it repairs 0.63–0.74— and the residual d repairs only 0.17–0.28.

The same control on third-party skills. We repeated the corrupted control on the clinical calculators, where the arm already exists in the released harness (numeric constants in the prose perturbed, wording, structure and executable tools untouched). That material is free-form and answered through a chain of thought, and there the corrupted skill is worth *nothing*: on 20 rescued items it scores 0.050, which is what no skill at all scores, against 0.950 for the gold skill. The synthetic and the real material therefore agree, and they

Table 5: Full-dataset rerun against the subset the main text cites, Qwen3-8B on MedCalc, layer 8. Recovery is Eq. 3; intervals are group bootstrap over calculators. *per-calc* is the main text's restriction, *per-item* the weaker one. Controls reach 0.12 under the group screen and 0.25 under the item screen; each arm uses its own matched baselines.

arm	full, per-calc (n = 135)	subset (n = 60)	full, per-item (n = 369)
transplant `real`	**0.89** [0.76, 0.97]	0.87	0.94
same skill, matched prefix	0.65 [0.37, 0.87]	0.73	0.84
half dose $\alpha = 0.5$	0.04	0.06	0.11
positions permuted	0.05	0.08	0.21
norm-matched noise	0.02	0.10	0.18
other calculator	0.05	—	0.25
other family, mean	0.05	0.06	0.16
same family, mean	0.12 [0.00, 0.25]	0.05	0.24
identity patch	0.00 [0.00, 0.00]	0.00	0.00
rank 128 / 64 / 32 / 16	0.77/0.63/0.12/0.05	0.77/0.54/0.13/0.06	0.88/0.70/0.22/0.18
bottom-64 / bottom-16	0.03/0.04	0.00/0.06	0.11/0.12
quarters $q_0/q_1/q_2/q_3$	0.09/**0.56**/0.12/0.05	0.12/0.56/0.12/0.12	0.18/0.64/0.22/0.19

Table 6: The same decomposition against three control documents. Synthetic control, multiple choice, Qwen3-1.7B, n = 358, rescued cell n_R = 82, every one of layers 21–27. "in context" is the control document actually placed in the prompt; "+g" is its state injected at the final position of the no-document prompt. That the two agree to within 0.03 for three very different documents is the strongest validity check we have on the patch itself. d is the same measurement in all three rows; only what it is subtracted against changes.

control document	fraction of the rescued cell correct		
	control in context	$h_{\text{no}} + g$	$h_{\text{no}} + d$
unrelated text, matched length	0.098	**0.098**	**0.500–0.805**
same document, lines scrambled	0.659	0.634–0.671	0.171–0.280
same document, factors replaced	0.720	0.720–0.744	0.171–0.256
gold skill in context	1.000	—	—
no document	0.000	—	—

agree on the thing that matters—whether corrupting a document costs you anything is a property of the answer format, not of the document.

Read as an attribution rather than as three competing estimates, this says something uncomfortable about the multiple-choice form of the task. On the full 358 items the gold document is worth 0.179 → 0.288; a scrambled version of it is worth 0.179 → 0.268 and a version with every factor wrong is worth 0.179 → 0.237. Four fifths of the measured skill effect survives destroying the order of the document, and just over half survives replacing every number in it. What the model needs, in this format, is mostly a document containing the right vocabulary and the right kinds of number—not a correct table. The correct table is worth the remaining 0.02–0.05 absolute, or 0.17–0.28 of the rescued cell.

That is a statement about the answer format as much as about the model. We registered the test before running it: in the free-form format the number has to be produced, so a corrupted table should be worth nothing. It is (Table 7). Corrupting every factor keeps 0.54 of the multiple-choice effect on the full item set and *none* of the free-form effect's 0.467. Scrambling the line order costs little in either, which is what it should: our shuffle permutes lines, and each row of the conversion table is one line, so every factor is still there to be retrieved—just not in a readable order.

C.1 Multiple choice manufactures a floor and merges retrieval with execution

Our synthetic items exist in two forms with identical content: four-option multiple choice, and free-form ("give the number only"). The forms do not measure the same thing, and there is a mechanism in the literature for why. Wiegreffe et al. [44] localise the choice of an answer *symbol* to a single middle layer and one to four attention heads per layer: in multiple choice the decision is assembled into one position and one token. That is exactly the regime in which a single-position patch transports the whole task, and it is why our final-position channel reads 1.000 there and 0.000 under a chain of thought on the same items.

Inspecting what the small model writes explains the top half. Without the document it echoes the quantity from the question ("2"); with the document it writes a *conversion factor from the table* ("21", "180", "60"). It retrieves and does not execute. Multiple choice scores retrieval as success whenever the retrieved number can be matched against an option, and the positional prior supplies a floor of 0.18–0.29 on top.

The bottom half is the sharper version, and it is a prediction we registered before running: if the multiple-choice form is solvable from a document's structure rather than its numbers, a table with every factor replaced must score near zero in free-form. It scores exactly zero.

Two consequences. A benchmark comparing skill effects across model scales in multiple-choice format is measuring a mixture whose composition changes with scale. And an ablation that perturbs a document and reports the accuracy drop is measuring, in multiple choice, mostly whether the document is still recognisably a document.

D THREE MEASUREMENT TRAPS, IN FULL

D.1 The aggregate gold-answer log-probability ranks the controls first

Averaged over all items, the log-probability of the gold answer under each patched condition at layer 27, the last, orders as

$$\text{mean vector } -2.85 \;>\; \text{filler state } -5.20 \;>\; \textbf{true skill state } -6.60 \;>\; \text{no document } -7.30 \;>\; \text{wrong skill } -10.29,$$

i.e. a content-free average vector scores 3.75 nats *better* than the state the model actually builds while reading the document, with accuracy 0.256 against that state's 0.436.

The cells of §4.1 account for the gap arithmetically (Table 16): F and B—where the document does not work—contribute 133% of it at layer 21, where the gap is 2.75 nats, while on R the true state wins by 2.5 nats and 0.69 in accuracy. The mechanism is visible in the option distribution: the mean vector raises the entropy over the four answer tokens from 0.187 to 0.972 nats (maximum $\ln 4 = 1.386$). A flattened distribution cannot assign any token −19 nats, so it looks excellent exactly where the true state is confidently wrong, and flattening never promotes a token to the argmax, so accuracy does not move. The one point the two channels agree on is that the *wrong skill* is worst on both—positive evidence for content specificity, since injecting the wrong content actively misleads.

The practical rule is that an aggregate log-probability over a mixed item set measures confidence calibration as much as it measures content, and that a decomposition by outcome cell is the cheapest way to tell which.

D.2 A band where the forward pass is damaged

Between layers 17 and 20 every patched condition improves together and the conditions become indistinguishable. Reading by cell isolates the cause: on the K cell—items the model already answers correctly—the true-state patch drops the gold log-probability from −1.6 at layer 16 to −7.1 at layer 18, while the filler-state patch, built over an equally long prompt, stays at −1.5. Any transfer or recovery number falling in 17–20 is reported with that band marked.

D.3 What the wrong-document control is actually worth

Holding the prompt's length, structure and the document slot's position fixed and changing only what sits in that slot gives an ordered comparison the usual binary—document or no document—cannot make. On the rescued items, with everything else identical ($n = 40$, intervals in brackets): no document at all 0.025 [0.000, 0.073]; non-clinical prose in the slot 0.250 [0.116, 0.384]; this skill's own paired neutral document 0.175 [0.057, 0.293]; one fixed wrong clinical skill 0.150 [0.039, 0.261]; the gold document 0.950 [0.882, 1.000].

Two things follow, one solid and one not. The solid one is that the wrong-document control is not the same object as the no-document control: any document in the slot moves these items off the floor, so a paper that reports "with the document" against "without" is measuring presence and content together, and the presence part is not negligible on a cell selected for failing without. That is the reason every arm in §4.3 is measured against a receiver that holds a document rather than against an empty slot.

The one that is not solid is the ordering. A clinical distractor costing more than irrelevant prose (0.150 against 0.250) is the direction we expected—a near miss should be worse than an obvious miss, and the three conditions fall in that order—but every pair of intervals overlaps at $n = 40$, so we report a direction and not a result. We flag it because the design that would settle it is cheap and this material affords it: these 55 skills contain graded near misses by construction (four different formulae for the corrected QT interval, two for the delta gap), so "how similar must a wrong document be before it hurts" can be measured on an ordered scale rather than a binary. We have not run that experiment.

E THE GEOMETRY AT THE FINAL POSITION, IN FULL

Table 10 says the effect is content: a document that is present but wrong is worth nothing. The internal picture at the last prompt position says the opposite, and it says so exactly where the document is being read. The cosine between the displacement caused by the skill and the displacement caused by an unrelated document of similar length is 0.93 at layer 0, 0.9475 at layer 4 and 0.91 at layer 10 on the synthetic control, falling to 0.35–0.41 only from layer 18, i.e. after the model has stopped reading (§4.7). On a second family the same comparison at layer 10 gives 0.9716 between two mutually exclusive skills against 0.9445/0.9555 between each of them and an unrelated document—a gap of 0.016–0.027 with bootstrap intervals of width 0.001, so it is real and it is negligible.

Read alone, that says the injection is not content-specific. Under Eq. 1 it says something narrower and correct: over the first half of the network $\|d\|$ is only 0.38–0.45 of $\|t\|$, so most of what a cosine between h_{skill} and h_{wrong} measures is shared, and the interesting quantity is the part that is not. Three facts follow, all in Figure 7 with the numbers in Appendix I. The content component is a minority of the displacement wherever the document is being read—under half of $\|t\|$ through layer 14. Through layer 16 $\|g\|$ actually exceeds $\|t\|$: the *wrong* document moves the state further than the right one, and the two components partly cancel, which is why the ratio recovers deeper without g shrinking. And two items' content vectors are far more aligned when they use the same row of the conversion table than when they do not, so whatever d encodes is about the document's content rather than about the item.

None of this is causal evidence, and the rest of the paper is about why it should not be read as any. A displacement can be large

and do nothing; a direction can differ systematically between documents and carry none of its effect (§4.5). We use the geometry only to say which quantity is worth intervening on, and then intervene.

E.1 The decomposition and the answer-length ladder, in full

We add each component back to the no-document state at the last prompt position and let the forward pass continue, $h' = h_{\text{no}} + \alpha v$ for $v \in \{d_i, g_i, t_i, \bar{d}, d_j\}$, against an upper reference that replaces the state outright with h_{skill}. The protocol needs a task with an exact zero floor and a matched negative control, so it is established on the synthetic tier and then carried to the clinical one; Appendix E.3 gives the full sweep, and this is the only place in the paper where the synthetic material carries an argument of its own.

Three results, and the third is the one the rest of the paper uses. **The presence component repairs nothing but is not inert**: injected alone it takes the rescued cell to 0.098, and putting the wrong document in the prompt also gives 0.098—it faithfully reproduces the behaviour of having *a* document, which is worse than having none. **The content component repairs most of the cell**, 0.805 against an upper reference of 1.000, having been dismissed by every geometric measurement as a small residual. **The patch reproduces the document's whole behavioural signature**: across the four cells, replacing the position with h_{skill} gives 1.000/0.000/1.000/0.000 and adding d_i gives 0.805/0.075/1.000/0.186—the same signature, attenuated. A patch that only ever helped would be consistent with a generic "try harder" effect; one that also breaks exactly the items the document breaks is carrying the document's decision.

Which control you subtract chooses what you measure. d is defined against a control document, and the choice is a measurement decision rather than a detail. Against an *unrelated* document of the same length it is topic, procedure and numbers together and repairs 0.805 of the rescued cell; against the *same* document with every number replaced it is the numbers alone and repairs 0.171–0.256 (layers 21–27); against the same document with its sentences shuffled, 0.171–0.280. The three differ by a factor of three to four, so a paper that reports "the content component" without naming its control has not specified the quantity. Appendix C gives the full comparison.

E.2 What a single position can carry is the answer, not the content

Single-position patching is the standard move in this literature, and Table 26 is a multiple-choice result, so it is worth asking what that one position is carrying. The same items exist in a free-form numeric form, where the answer is a number the model must compute rather than an option it must select; there the skill is worth 0.006 → 0.332 and the rescued cell has 119 of the 358 items.

Two things change and one does not. The ceiling collapses: through layer 22 no condition exceeds 0.008, including replacing the position outright with the true skill state, and the best any single-position intervention achieves is 0.235, at layer 30. What does not change is the ordering—the content component recovers 0.210 and the presence component 0.008—so the conclusion of §4.2 survives the format while the magnitude does not (full sweep in Appendix I).

It is the format, not the model. The identical sweep on the *same model and the same* 358 *items* in multiple-choice form (Qwen3-8B, $n_R = 94$) recovers 0.957 at layer 26 and 1.000 at layers 29, 30 and 35. Same model, same items, same intervention, same layers: 1.000 when the answer is one option letter, 0.235 when it is a number the model has to produce.

The reason is mechanical and can be counted. The gold answers run one to five tokens under this tokenizer and 341 of the 358 use two to four. A patch at one prompt position can fix at most the first generated token; the rest are produced in a context that still does not contain the conversion table. Multiple choice compresses the answer to exactly one token—which is exactly the unit a single-position patch transports. Patching the last k positions instead of one does *not* lift the ceiling in proportion: $k = 1, 4, 16$ recover 0.235, 0.261 and 0.403, and $k = 32, 48, 61$ recover 0.403, 0.395 and 0.370, so sixteen positions carry about half of what the document carries, four—the chat suffix—carry nothing extra, and everything past sixteen carries nothing at all.

On third-party skills the same probe reads nothing. There, replacing the final prompt position with the with-skill state repairs at most 3 of 18 rescued items (layer 10) and 0–1 from layer 14 on and transplanting the question span—50 to 1697 positions, into a length-matched receiver—repairs nothing beyond the receiver's own baseline at any of seven layers (Table 9), with the self-transplant arm on that baseline to the item, so the instrument is live and the null is real. §4.3 shows what does work on the same material.

Answer length is the cause, and that is a controlled comparison. The step from the synthetic tier to the clinical one changes the skill, the task, the prompt length and the fraction of it a patch covers, so it cannot on its own identify what makes the injection fail. We therefore made answer length the only thing that moves: the *same* 80 synthetic items, the same model, the same document, prompted to reason step by step and give the answer on a final `ANSWER:` line. In that format the skill is worth 0.000 → 1.000—every item is rescued, $F = K = B = 0$—so there is more effect to recover than in either other format, and the rescued cell is the whole item set.

The single-position patch recovers **0.000** of it (§4.2), as do d and g. The last-layer instrument check passes on 10 of 10 items in the same run.

The range over which "a skill's effect is carried by a vector" is true is the range over which the answer is short—and the clinical result is then not a property of that material but an instance of this.

It is a ceiling, and more positions do not raise it. Patching one position is a choice, not a measurement, so we repeated the sweep patching the last k prompt positions for $k \in \{1, 4, 16, 32, 48, 61\}$, every one of the 36 layers. The no-document prompt is 61–63 tokens, so $k = 61$ writes all of it and the ladder has no further rung. Recovery on the rescued cell, at each k's best layer, is 0.235, 0.261, 0.403, 0.403, 0.395 and 0.370, and 0.403 is also the largest value over every k and every layer. Four positions buy nothing—the last four tokens are the chat suffix—sixteen, which reach back into the

question, roughly double the recovery, and past sixteen nothing is bought: at $k = 32$ the items answered correctly are item-for-item those answered correctly at $k = 16$, with no discordant pair, and at $k = 61$ four are lost and none gained (exact McNemar $p = 0.125$). We expected the ladder to keep climbing and registered that prediction before the larger k ran; it does not climb.

The ceiling is therefore not a budget, and it is not an artefact of writing a long prompt's states into a short one: the identical probe, receiver and donor recover 1.000 on the rescued cell of the multiple-choice format, where the answer is one token at one position. What separates 0.403 from 1.000 is the answer format.

It is *not* explained by the patched positions being ones the document does not occupy. On this same free-form material the document's own 688 positions do no better: they recover 0.408 at layer 0, which is close to putting the document back, and then at most 0.17 from layer 3 and nothing from layer 15 (Table 8), below what the last sixteen positions reach. In free-form no channel on this task exceeds about 0.41 at any layer, so the ceiling is a property of the answer format rather than of which positions are written. The comparison that does isolate position is on the clinical material, where receiver, items and layer are held fixed: the document's own span recovers 0.89 and up to 256 positions after it recover 0.05 (§4.2).

One thing this ladder does not separate is the receiver. It patches into the *no-document* prompt, while span injection patches into a length-matched *wrong-document* one, so a receiver effect is not excluded here. Where we have measured it—the same decomposition at the final position on the synthetic multiple-choice task—the wrong-document receiver is never worse than the no-document one and is better in the middle layers (Table 26), so it would move the ceiling the wrong way; but we have not run it in free-form. We therefore read the low ceiling as a scope condition on single-position and short-suffix patching under a multi-token answer format, rather than as a negative result about d. In a multiple-choice item the answer *is* one token at one position, so a patch that transports the model's decision transports the whole task and looks like it transported the document. What the final prompt position holds by layer 21 is the model's *answer*, not the document's content—consistent with the knockout result that reading the document is finished by that layer (§4.7).

E.3 The decomposition on the synthetic tier, in full

The geometry at that position says the opposite of the behaviour. Appendix E gives that measurement and why it cannot settle the question on its own. We use geometry only to choose what to intervene on, and then intervene.

We add each component back to the no-document state at the last prompt position and let the forward pass continue: $h' = h_{\text{no}} + \alpha v$ for $v \in \{d_i, g_i, t_i, \bar{d}, d_j\}$. The upper reference is replacing the state outright with h_{skill}; the lower reference is no intervention.

Figure 5 is the causal counterpart of that geometry (numbers in Table 26, Appendix I). Four of its curves are worth reading separately.

The presence component repairs nothing—0.098, eight items of 82, at every layer past 20, and 0.00–0.02 before. It is not inert noise: injected on its own it takes whole-set accuracy from the no-document baseline of 0.180 to 0.151, which is the accuracy of actually putting the wrong document in the prompt. On the rescued cell it gives 0.098, and the wrong document in context also gives 0.098. It faithfully reproduces the behaviour of having a document present, and that behaviour is worse than having none.

The content component repairs most of the cell—0.805 at layer 26 against an upper reference of 1.000, having been dismissed by every geometric measurement in §E as a small residual.

At one patched position it is under-scaled, not incomplete. The dose-response is monotone and steep—$\alpha = 0.5, 1, 2$ give 0.31, 0.81, 0.98—so the missing 0.19 at $\alpha = 1$ is a magnitude deficit along a correct direction, which is expected: $h_{\text{wrong}} + d_i$ is exactly h_{skill}, while adding d_i to h_{no} lands elsewhere. It does not generalise to more positions: at $k = 16$ in free-form the dose is flat between $\alpha = 1$ and 2 (0.232, 0.268) and never approaches the full replacement's 0.554. The two components are additive by construction and not additive in effect. What survives every configuration is the asymmetry: g alone is inert at every k, layer and α.

The receiver does not matter, and a single shared vector is not enough. Injecting h_{skill} into the prompt that already holds the *wrong* document gives 0.979–1.000, the same as into a prompt with no document—whatever the surrounding 700 tokens contribute, the final position does not need it. But $\bar{d}$, one vector for the whole item set, which is what "a reusable skill vector" would mean, recovers 0.049–0.122 from layer 21 against 0.500–0.805 for the item's own. Splitting d_i along $u = \bar{d}/\|\bar{d}\|$ and orthogonal to it and injecting each half separates them and reverses the answer at layer 21: past it the parallel half carries ≈ 0.9 of the norm and 0.05 of the effect while the orthogonal half carries 0.44 of the norm and 0.60 of the 0.81; before it the two swap, inside the band §4.2 independently identifies as one where the forward pass is disturbed.

The patch reproduces the document's whole behavioural signature, not just its successes. Across all four cells, replacing the final position with h_{skill} gives 1.000/0.000/1.000/0.000 on rescued / persistent / kept / broken at layer 26 on the 1.7B, and 1.000/0.024/1.000/0.000 at layer 30 on the 8B—cell by cell, what the document itself does by the definition of the cells. It reproduces the skill's failures and its damage as faithfully as its repairs. Adding d_i gives 0.805/0.075/1.000/0.186 and 0.864/0.024/1.000/0.077: the same signature, attenuated. A patch that only ever helped would be consistent with a generic "try harder" effect; one that also breaks exactly the items the document breaks is carrying the document's decision. The presence component passes the matching test against *its* document: $h_{\text{no}} + g$ scores 0.098 on the rescued cell where the wrong document in context scores 0.098 (1.7B), and 0.250 where it scores 0.250 (8B), across four different control documents (Appendix C).

F THE SPAN BATTERY, IN FULL

§4.2 is a negative result about a probe, not about a model. The probe fails because one position cannot hold a decision that occupies hundreds of tokens. The obvious repair is to stop using one position, and the natural place to use instead is the one group of positions whose size grows with the document: the document's own tokens.

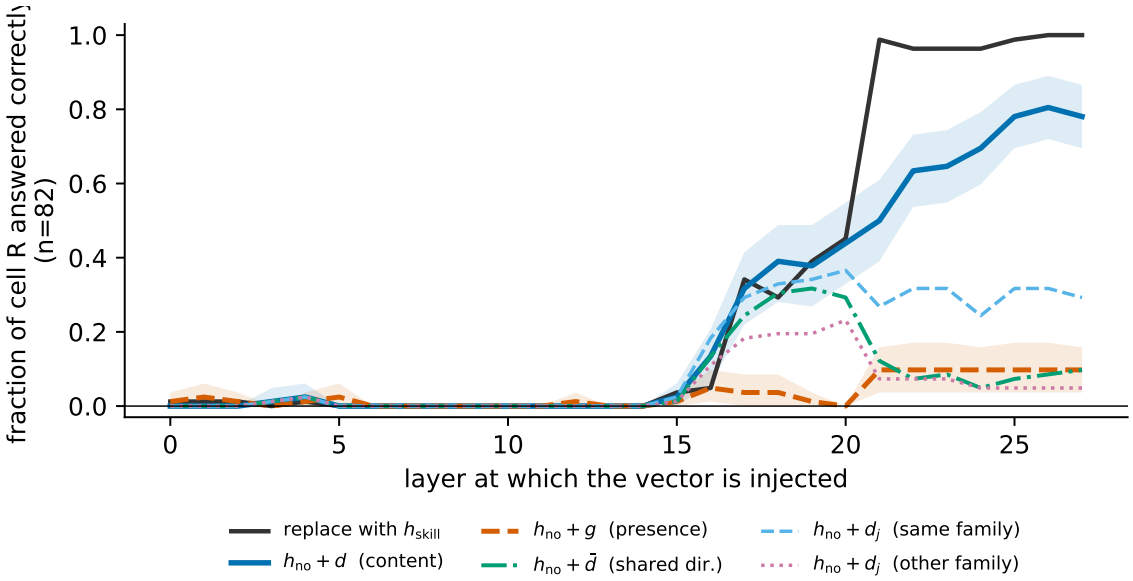


Figure 5: What each injected component repairs, on the cell the document rescues. Shaded bands are bootstrap 95% CIs for the two conditions the argument rests on. $h_{no} + t$ is omitted because it equals h_{skill} identically and its curve lies under the black one—we run it, and it agrees item by item. Synthetic control, multiple choice, Qwen3-1.7B, fp32, $n = 358$, rescued cell $n_R = 82$.

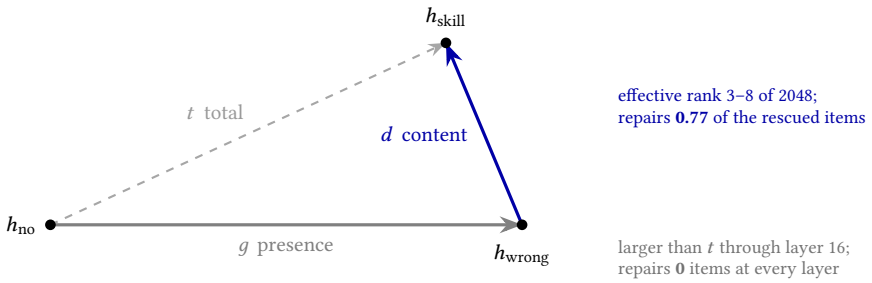


Figure 6: The decomposition, drawn to scale at layer 10 of the synthetic control ($\|g\| = 39.0$, $\|d\| = 16.3$, $\|t\| = 36.0$). A document of the right length and the wrong content already moves the final prompt position most of the way—in fact slightly further than the real skill does, which is why the content arrow points partly back. Everything this paper is about is the short blue arrow.

Table 7: The same items under two answer formats. Top: the effect itself. In multiple choice the small model shows a +11pp skill effect over an 0.18 floor; in free-form it shows nothing at all, over a floor of exactly zero, and the effect appears only at 8B. Top block and the multiple-choice column of the bottom block are the 358-item reruns; the free-form control column is the earlier 120-item sample, which has not been rerun with the corrupted and scrambled documents. Bottom: the same four documents under both formats, where "share" is (acc − none)/(gold − none). Destroying every number in the document keeps just over half of the effect in multiple choice and none of it in free-form.

model	format	no document	wrong document	gold skill	effect
Qwen3-1.7B	multiple choice	0.179	0.151	0.288	+10.9pp
Qwen3-1.7B	free-form	**0.000**	—	**0.000**	**0**
Qwen3-8B	multiple choice	0.274	0.249	0.425	+15.1pp
Qwen3-8B	free-form	**0.006**	**0.006**	**0.332**	**+32.6pp**

document in context	multiple choice (1.7B)		free-form (8B)	
	acc	share	acc	share
none	0.179	—	0.000	—
same document, *factors* replaced	0.237	0.54	**0.000**	**0.00**
same document, *lines* scrambled	0.268	0.82	0.325	0.70
gold skill	0.288	1.00	0.467	1.00

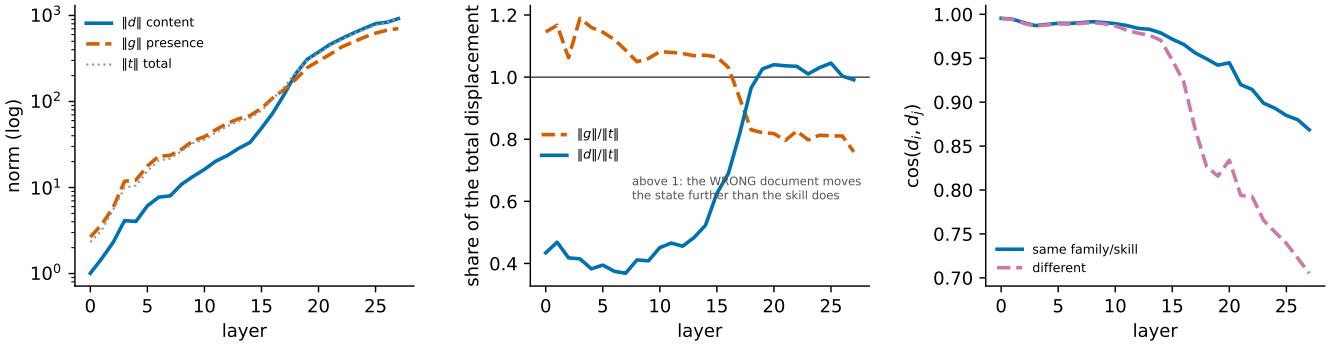


Figure 7: The decomposition across depth, synthetic control, Qwen3-1.7B, fp32, $n = 358$. Left: the three norms. Middle: each component as a share of the total displacement; above 1 the wrong document moves the state further than the skill does, which is the case until layer 17. Right: how aligned two items' content vectors are, split by whether they use the same row of the conversion table.

Table 8: Peak accuracy of each transplant channel, by model and answer format. "Donor" is the with-skill forward and "receiver" the prompt being patched into; the two span channels patch into a length-matched filler-substituted prompt, the final-position channel into the no-document prompt, which is why their baselines differ. The final-position rows are the 358-item reruns; the two span channels are the earlier 39-item transplant runs, which were not rerun. Peaks in bold where the channel recovers most of the donor.

channel	configuration	recv.	peak (layers)	donor
document span (688 tok)	1.7B, multiple choice	0.154	**0.436** (3–4)	0.436
	8B, multiple choice	0.233	**0.575** (0–6)	0.550
	8B, free-form	0.008	0.408 (0 only)	0.467
question span (26–50 tok)	1.7B, multiple choice	0.154	0.333 (12–14)	0.436
	8B, multiple choice	0.233	**0.517** (21)	0.550
	8B, free-form	0.008	0.025 (any, i.e. none)	0.467
final position(s)	1.7B, MC, rescued cell	0.000	**1.000** (26–27)	1.000
	8B, MC, rescued cell	0.000	**1.000** (29–35)	1.000
	8B, free-form, rescued cell	0.000	0.235 (k=1), 0.403 (k=16)	1.000

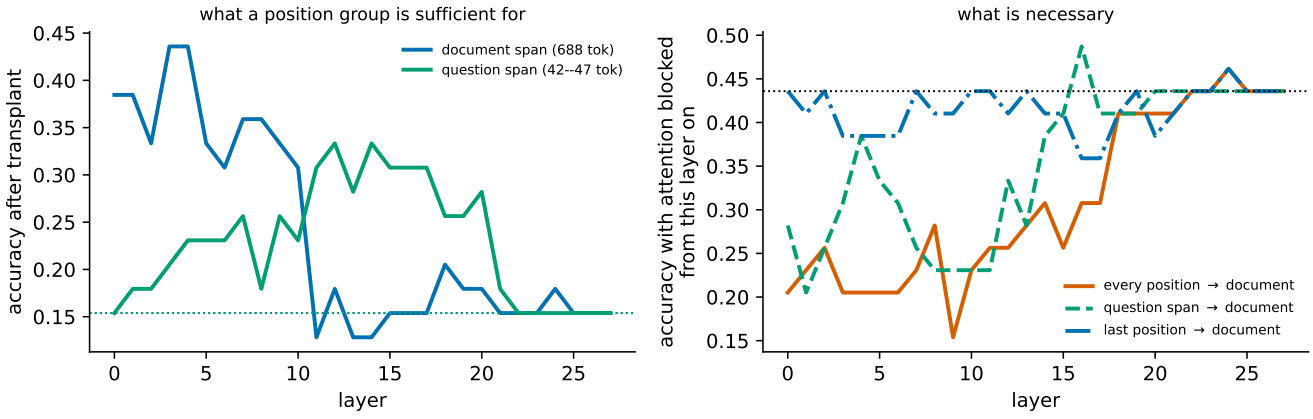


Figure 8: Left: what each position group is sufficient for, as a function of the layer at which its states are transplanted; dotted lines are each channel's own receiver baseline. Right: what is necessary—accuracy when attention to the document is masked from the given layer to the end, cumulatively; the dotted line is the unmasked baseline. Synthetic control, multiple choice, Qwen3-1.7B, $n = 39$.

Table 9: Question-span transplants on the same material and construction. *Which* positions are transplanted is the whole difference between this table and Table 11.

transplanted positions	donor	L0–12	L16	L22	L30
question span (50–1697 pos., $n = 40$)	the gold skill	0.10–0.23	0.150	0.200	0.154
	the receiver itself (no-op)	0.13–0.20	0.175	0.125	0.154
no document / the gold skill in context		0.025 / 0.950			

The receiver. The donor is the with-document forward; the receiver is the *same prompt* with the document's tokens overwritten, one for one, by the tokens of *one fixed clinical skill document*—the same document for every instance and every skill, and the gold document of none of them.

Both halves of that choice were forced by measurement. It has to be fixed, because a receiver built from each skill's own paired control makes the receiver's states depend on the instance, and then every arm of the form $h_{\text{recv}} + (\cdot)$ inherits an instance-dependent floor; measured, the with-document states vary across a skill's instances by a relative 3.5×10^{-3} while such a receiver's vary by 0.36. We use a *clinical* document rather than neutral prose because it is the same condition the behavioural runs call "wrong skill", so the receiver's baseline means what the paper's presence effect means; §D.3 measures the two side by side and finds them close (0.150 against 0.250, $n = 40$, overlapping intervals).

Length-matching is not a refinement here either, it is the whole experiment. An earlier version of this run used the no-document prompt, which is some 900 tokens shorter, and aligned the spans by their ends; the real arm and the wrong-document arm then both read 0.200—the signature of two arms broken in the same way by a rotary phase shift, which is easy to read as an absent effect. With a matched receiver every position keeps its index, the two sequences are identical outside the span, and the question span is verified token-identical between donor and receiver before anything is patched.

Because the receiver already holds a document, the substitution obeys $h_{\text{recv}}(\text{span}) + d(\text{span}) = h_{\text{skill}}(\text{span})$: it is $h_{\text{recv}} + d$ at $\alpha = 1$. That is what makes the rest of this section possible.

It carries, completely. Figure 1 puts the two channels side by side on the same model and the same decomposition. At the last prompt position, five arms—including replacing the entire state with the with-document one—sit between 0.050 and 0.150 against a no-document baseline of 0.050. Over the document's own span, after the instrument fix, the corresponding span content matrix gives 1.00 against a receiver at 0.00 ($n = 16$, four calculators).

Before the fix the same panel read 0.950 on twenty items, $\rho = 1.06$: the transplant and the document disagreed on one item and the transplant got it right. The honest reading of every ρ near or above 1 here is "indistinguishable from having the document in the prompt".

How deep the handover works (before the fix). Transplanting the *document's own* states over the same construction recovers the document's entire effect through layer 12 and then falls off a cliff (Table 11). Layer 0 is close to tautological—a layer's output over the document's tokens is not far from putting those tokens back—so the load-bearing entries are layers 8 and 12, a third of the way into the network, where the receiver has already processed eight or twelve blocks' worth of filler in those positions and the transplant still recovers everything. Past layer 14 a single layer's worth stops being enough: $\rho = 0.14$ at 15 and 16, and 0.00 at 24 and 30.

Writing the span at *every* layer at once returns it to $\rho = 1.00$, so what fails past layer 14 is the handover at one depth, not the mechanism.

The profile replicates; its one anomaly does not. We ran the sweep a second time on the same items against the other receiver—the single fixed wrong skill rather than each skill's paired control—and the two agree wherever the first is well behaved: +1.00 against +1.00 at layer 0, +1.06 against +1.07 at 4, +1.03 against +1.14 at 8, +0.81 against +1.00 at 12 (Table 19). The single bump in the first run, $\rho = 0.07$ at layer 18 against 0.29 at 20, does not survive: the replication gives +0.16 at layer 20, below its own layer 16. We read the profile and not the individual layers, and we read that bump as noise.

The surviving window is a window in *relative* depth: layers 0–12 of 36 here, 0–10 of 28 on the synthetic tier, both about a third of the way in.

It behaves like a mechanism, not like a perturbation. Four controls, all at layer 8, all in Figure 1, and all read on the calculators a wrong document never rescues.

Dose: after the fix, $\alpha = 0.4, 0.5, 0.55, 0.6, 0.7, 1, 2$ give $\rho = 0.06, 0.00, 0.06, 0.62, 0.94, 1.00, 0.81$ ($n = 16$). The transition is a step and not a ramp—half the content is no better than none of it, since the state is then a midpoint between two coherent documents and corresponds to neither—and overshooting costs a little. An injection that merely had to be large would show neither. Before the fix the same sweep read 0.17, 0.03, 0.17, 0.88, 0.97, 1.00 ($n = 30$), and a coarser one 0.06, 1.06, 1.06, 0.89 at $\alpha = 0.5, 0.75, 1, 2$ with the per-position rescaled doses coinciding (0.06 and 0.89 both ways).

Transfer: after the fix, another skill's vector over the prefix the two documents share gives 0.19 against 0.94 for this skill's own vector on the same positions ($n = 16$), and on forty calculators a donor from another family 0.06 and one from the same family 0.06 against 0.73. Before the fix, the mean content vector of a *different* skill, written over the prefix the two documents share, gave $\rho = 0.06$ against 1.00 for this document's own vector written over *the same positions*. Because every skill's vector is taken against the same fixed control, this arm is exactly that skill's own span states rather than a hybrid, so it is the cleanest form the transfer test can take; and the paired control is what makes it a statement about content rather than about how many positions were patched.

Order: the same vectors, permuted across the span's own positions, give $\rho = 0.00$ ($n = 16$) and 0.08 on forty calculators after the fix (0.12 before). The span is not a bag—which position holds which part of the document is most of the effect.

Norm: Gaussian noise rescaled to each position's $\|d\|$ gives $\rho = 0.00$, exactly the receiver (0.10 on forty calculators). The effect is not a consequence of writing something large into those positions.

The split these four are read on, and why it is not cherry-picking. On four of the ten calculators the receiver—a wrong clinical document—already solves the items. Before the fix every span arm worked there: a related-skill donor scored $\rho = 0.71$, an unrelated one 0.43 and norm-matched *noise* 0.14, against 0.00, 0.06 and 0.00 on the five where the receiver scores zero, and pooling the two groups turned a null transfer arm into an apparent $\rho = 0.38$. After the fix six calculators fall on that side; noise no longer works there (0.00, $n = 24$) but another skill's vector still reads 0.47 against 0.19 on the four restricted calculators: there the wrong document in the slot was leading the model astray, and a coherent replacement lets it fall back on a formula it already knows.

Table 10: Behaviour on the real skills, Qwen3-8B, all three arms on the same 1,100 items, CIs bootstrapped over the 55 calculators. The presence effect (wrong skill minus no skill) is a clean zero, so the +39.7pp is content.

	n	accuracy	95% CI
gold skill	1100	74.2%	[65.8, 81.8]
no skill	1100	34.5%	[26.5, 42.8]
wrong skill (length-matched)	1100	31.5%	[23.7, 39.5]
gold − no skill	1100	+39.7pp	[+31.7, +47.7]
gold − wrong skill	1100	+42.6pp	[+34.7, +50.5]
presence: wrong − no skill	1100	**−2.9pp**	[−8.8, +3.0]

The split conditions on the *receiver*, which is a control, and on no intervention arm, so it cannot manufacture the contrast it is used to report; and it is taken per calculator rather than per instance because "this model can do this calculator unaided" is a property of the calculator. Table 20 gives every arm on all forty items for comparison.

It replicates on four times as many skills. The ten calculators above are the ten with the most rescued instances, and twenty items from five calculators is the single most obvious reason to stop reading. We therefore ran the battery on *forty* calculators, recomputing the restriction inside the larger set rather than carrying it over. Of the forty, fifteen are calculators the wrong document never rescues, which leaves 60 items—three times the original—and every arm holds (Table 2): the document's own vector recovers $\rho = 0.87$, the same vector over only the prefix another skill's document shares 0.73, half a dose 0.06, a donor from another family 0.06, permutation of the span's own positions 0.08, and norm-matched noise 0.10, against a null arm at exactly 0.00 and a receiver at exactly 0.000, and a donor from the same family 0.06 (rerun after the instrument fix). The separation is between 0.87 and at most 0.10, on three times the material, with item-level intervals as identified in the main-text caption.

And it fails where the document fails (before the fix). On the persistent-failure cell—items the gold document does *not* rescue—the same transplant at the same layer gives 0.100, equal to the receiver's 0.100, against the document's own 0.025 ($n = 40$, ten calculators; the null arm read 0.150 in this run). The channel transports what the document does, including doing nothing. An arm that only ever helped would be consistent with a generic "try harder" effect; this one is not.

Why it can carry when a single position cannot. The span is 415 to 1,630 positions wide on the forty calculators, the same order as the answer it has to support. This is the positive half of §4.2's capacity statement, and the two together give the rule the paper is built on: *a patch transports what fits in the positions it is given.* A one-token answer fits at one position; a chain of thought fits nowhere except on a group of positions as large as the document itself.

Table 11: The document-span transplant by layer. Qwen3-8B, 55 third-party clinical skills, the 40 rescued items, the full chain of thought decoded and graded by the task's own scorer. The receiver is the with-skill prompt with the document's tokens replaced by a filler document, before the instrument fix, so it is the same length and every transplanted position keeps its rotary phase. Restricted, as everywhere in this section, to the calculators the receiver never solves. This sweep used each skill's paired control as the filler and the battery of §4.3 the single fixed one; over all forty items the three baselines agree to within 0.01 across the two (0.175/0.025/0.950 against 0.167/0.028/0.944), so the profile does not depend on that choice—the unrestricted version is Table 18. The no-op arm—the receiver's own states written back into the receiver—is reported wherever it was run and must equal the receiver's baseline; it does, at every layer.

layer	accuracy	CI_{95}	fraction of the document's own effect
0	0.875	[0.713, 1.000]	+1.00 (no-op arm 0.000)
4	0.938	[0.819, 1.000]	+1.07
8	1.000	[1.000, 1.000]	+1.14
12	0.875	[0.713, 1.000]	+1.00 (no-op arm 0.000)
13	0.625	[0.388, 0.862]	+0.71 (no-op arm 0.000)
14	0.688	[0.460, 0.915]	+0.79 (no-op arm 0.000)
15	0.125	[0.000, 0.287]	+0.14 (no-op arm 0.000)
16	0.125	[0.000, 0.287]	+0.14 (no-op arm 0.000)
18	0.062	[0.000, 0.181]	+0.07 (no-op arm 0.000)
20	0.250	[0.038, 0.462]	+0.29
24	0.000	[0.000, 0.000]	+0.00
30	0.000	[0.000, 0.000]	+0.00 (no-op arm 0.000)
every layer	0.875	[0.713, 1.000]	+1.00
the filler document, unpatched	0.000		0.00
the gold document in context	0.875		1.00

Table 12: Geometry of the content component over the document's own span, by layer, Qwen3-8B, $n = 40$, all against the same fixed control document. Unlike the same quantity at the final prompt position, this vector is the size of the state it displaces. Two skills' vectors are substantially aligned and most of each one's energy lies along the all-skill mean—which, injected alone, recovers 0.11 here and 0.06 with the document after the question (both pre-fix) (§F.1). Participation ratios are centred, so they count the directions the span's positions *differ* in. Cosines between content vectors depend on which control was subtracted: against each skill's own paired control the cross-skill cosine reads 0.10 instead of 0.54, which is a fact about the control and not about the skills.

layer	$\|d\|$	$\cos(d_i, \bar{d})$	energy along $\bar{d}$	$\|d\|/\|h\|$	$PR(d)$	$PR(h_{skill})$	$\cos(d_i, d_k)$, other skill
0	10	0.720	0.54	0.91	81.8	50.5	0.483
4	28	0.738	0.56	0.94	77.1	45.7	0.511
8	53	0.755	0.58	1.04	94.8	63.8	0.539
12	68	0.766	0.60	1.09	64.2	46.3	0.560
16	106	0.759	0.59	1.19	17.1	13.9	0.556
20	153	0.755	0.58	1.15	17.5	15.2	0.547
24	266	0.750	0.57	1.06	17.2	14.8	0.535
28	455	0.756	0.58	1.08	14.4	13.1	0.547
32	751	0.753	0.58	0.98	17.5	13.8	0.541

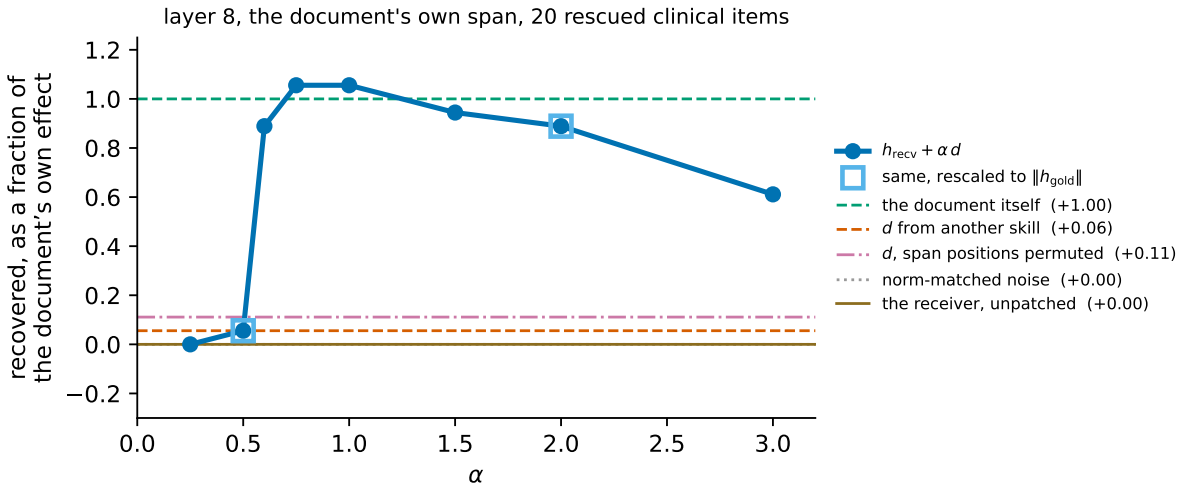


Figure 9: Dose response of the content injection over the document's own span, layer 8, with its controls. Open squares are the same dose rescaled per position to the with-document norm; they coincide with the plain arms. Before the instrument fix; the post-fix dose sweep and norm comparison are in Appendix F.1. The horizontal lines are the three controls and the two baselines.

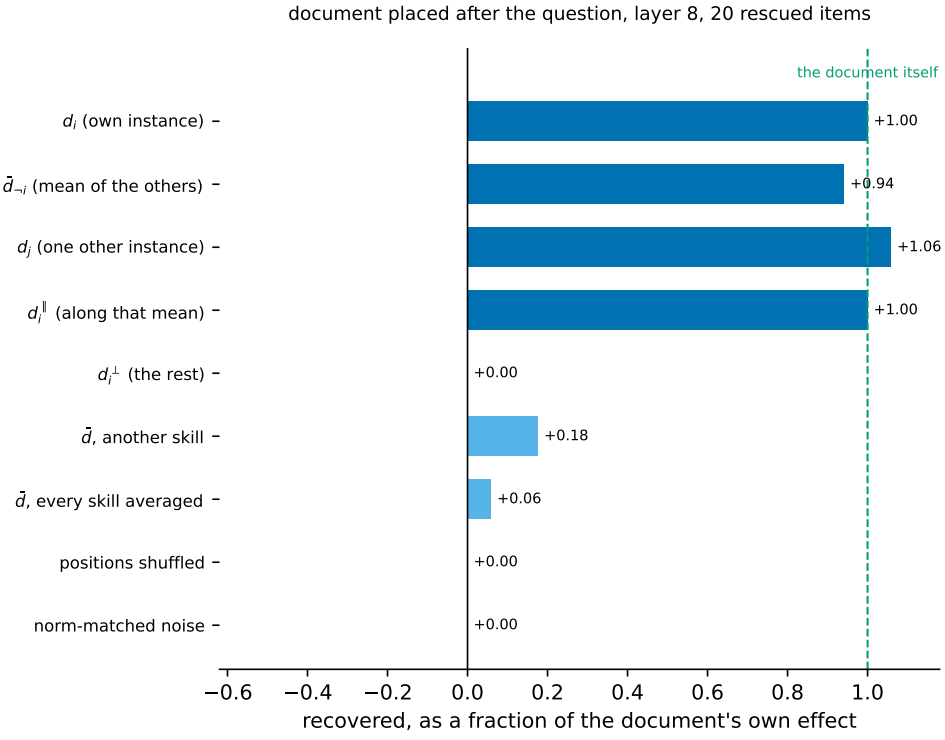


Figure 10: What the content vector contains once it is able to contain the task. Document placed after the question, layer 8, Qwen3-8B, the 40 rescued items, plotted as the fraction ρ of Eq. 3, before the instrument fix (post-fix numbers in Appendix F.1). Exact accuracies and intervals are in Table 21, Appendix I.

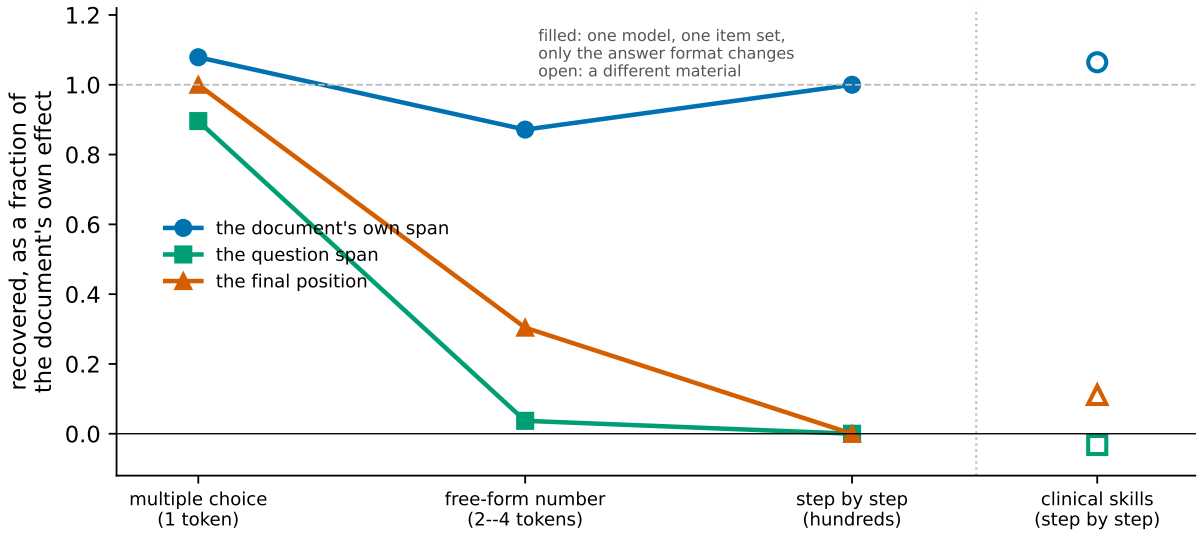


Figure 11: What each transplant channel recovers, as a fraction of what the document itself is worth in that setting, against how many tokens the answer occupies. Filled markers joined by lines are one model, one item set and one document with only the answer format changed; the clinical results are open and deliberately not joined, because that material changes the skill, the task and the prompt length at once. Plotting fractions rather than raw accuracies is necessary here: the four settings have donors at 0.550, 0.467, 1.000 and 0.950, and comparing raw numbers across them is how an earlier draft came to report that a *longer* answer made the document-span channel stronger. The document's own span is flat at 1.00 across every answer format. The other two channels collapse as the answer lengthens, and the clinical points land where the step-by-step column says they should.

F.1 What the carrying vector is made of, in full

The vector at the document's span reproduces the document's whole effect, so what it is composed of is answerable rather than speculative. Four decompositions, all at layer 8. **Unless marked "after the fix", the numbers in this subsection come from the ten-calculator development runs made before the instrument fix**, read against the receiver's 0.175 and the document's 0.950; where a rerun exists it is given and takes precedence.

Not a common shift. Averaging d over *every* skill in the set and writing that average gives $\rho = 0.11$, against 1.06 for the document's own. What makes this sharp rather than merely negative is how much of the vector that average is: measured against the same fixed control, two skills' content vectors have cosine 0.48–0.56 at every depth, the cosine to the all-skill mean is 0.72–0.76, and 54–60% of each d's energy lies along that mean (Table 12). *More than half of the content vector points in a direction that carries nothing.* This is the span-level counterpart of g's flat 0.098 at the final position in §4.2: presence is not what acts, in either place.

Not a magnitude. The informative test is the dose curve against its norm-matched twin. At $\alpha = 2$ the patched state is $2h_{\text{skill}} - h_{\text{recv}}$, which is 1.76× as long as the with-document state at layer 8; rescaling it per position back to $\|h_{\text{skill}}\|$ is a 43% change and costs nothing—$\rho = 0.89$ either way. At $\alpha = 0.5$ the state is 0.88× and rescaling is likewise free (0.06 both).

We also ran the reverse, rescaling the with-document states down to the *receiver's* norm, and it changes nothing ($\rho = 1.06$)—but that arm is uninformative and we report it only to say so: the two norms differ by 3.5% at layer 8, so it is barely a manipulation. The $\alpha = 2$ comparison is the one that carries the claim. The dose curve and its norm-matched twin coincide at every α we ran (0.5: 0.350 vs 0.350; 2: 0.875 vs 0.875; 3: 0.675 vs 0.650). The document's effect is carried by the direction its span states point in and not by how far along it they go. Norm-matched Gaussian noise, correspondingly, gives 0.100, below the receiver. The norm arms were not rerun after the fix. (The arm listed next to the transplant in Table 2, 0.73, is not a norm manipulation: it is the transfer test's paired control, the document's own d written over only the prefix another skill's document shares.)

Graded in α, and the transition is a step. After the fix, on the ten development calculators ($n = 16$), $\alpha = 0.4, 0.5, 0.55, 0.6, 0.7, 1, 2$ give $\rho = 0.06, 0.00, 0.06, 0.62, 0.94, 1.00, 0.81$: at the receiver up to 0.55 and at the document by 0.7. Before the fix the same sweep read 0.17, 0.03, 0.17, 0.88, 0.97, 1.00 ($n = 30$ under a looser restriction), and an earlier receiver gave $-0.14, +1.03, +1.03, +0.86$ at $\alpha = 0.5$, 0.75, 1, 2 (Figure 9, pre-fix): half the content is no better than none of it, while three quarters is already most of the effect. Past $\alpha = 1$ the recovery decays slowly rather than breaking, and against the other receiver, where the fine grid was run, it decays the same way with the norm held fixed ($\alpha = 2$: 0.875 plain against 0.875 rescaled;

$\alpha = 3$: 0.675 against 0.650), so what overshooting costs is not the size of the state.

Localised, and in the section one would guess. Writing d over only part of the span localises the effect within the document without changing a single prompt token—the length, the positions and the rotary phases are all untouched, and only which states are overwritten changes. By halves (pre-fix), the first gives $\rho = 1.06$ and the second 0.28, and the first half agrees with the whole span item by item on 18 of 20. By quarters:

quarter of the span	n	1st	**2nd**	3rd	4th
forty calculators, after the fix	60	0.12	**0.56**	0.12	0.12
ten calculators, after the fix	16	0.12	**0.81**	0.06	0.06
ten calculators, before the fix	20	0.28	**0.83**	0.11	0.44
controls on the same forty (Table 2)	60	half dose 0.06, permuted 0.08, noise 0.10			

These documents make that interpretable, because all ten have the same section order: title, *Required Inputs*, *Computation* (sometimes preceded by a *Unit Conversion*), *Calculation Tool*, *Example*. Measured in the units the intervention actually uses—token positions within the span, not characters—the computational section's *heading* falls at 0.18–0.42 of the span, *Calculation Tool* at 0.47–0.70, and *Example* at 0.61–0.79. So the second quarter is the *body* of the computational section, the formula itself, for every one of the ten; the first quarter is the title, the input list and at most the opening of that section; the third is a Python function signature and the start of a worked example; the fourth is the rest of the example.

Read that way the procedural core carries most of the effect, and every other quarter alone is at the control floor. Two things changed with the fix and the larger set. The worked example, which read 0.44 before the fix, reads 0.06 and 0.12 after it: *we withdraw the earlier statement that it carries about half as much as the formula.* And the formula alone, 0.81 on ten calculators, is 0.56 on forty against the whole document's 0.87, so on the larger set the other three quarters matter in combination although none matters alone. The boundaries are approximate and vary by calculator, so we report the quarters and the section map rather than claiming an exact alignment.

Permuting the span's own positions, which preserves the multiset of vectors and destroys only the correspondence between them and the tokens, gives 0.08 on forty calculators after the fix (0.21 before it): nearly all of the effect is in that correspondence.

Task information, when the vector is allowed to have any. Under the prompt format used everywhere else in this paper the question does not arise: the document precedes the question, so its states are a function of the document alone and we verified that they agree across a skill's instances (bit for bit at equal prompt length, cosine ≥ 0.997 otherwise). To ask what happens when the vector *can* carry the task, we move the document behind the question, keeping the answer-format instruction last, and pad each instance's question with filler tokens so that the document begins at the same absolute index for every instance of a skill—without which a donor from another instance would be written at the wrong rotary phase, the error described in §4.3. The format costs almost nothing behaviourally (the document is worth $0.025 \rightarrow 0.925$ instead of $0.025 \rightarrow 0.950$) and it raises the receiver's own baseline to 0.350.

In that format the vector *can* carry the task, and at layer 8 it does not use it (read, like everything else here, on the calculators a wrong document never rescues). After the fix ($n = 28$, seven calculators) the instance's own vector recovers $\rho = 1.00$, the mean of the skill's other instances' vectors 1.00, a single other instance's vector 0.96, and the orthogonal remainder—everything specific to this patient note—0.04. Before the fix (Figure 10) the same arms read 1.00, 1.00, 1.06 and 0.00. At layer 16 the picture changes: the instance's own vector still recovers 0.85, but the mean of the other instances' vectors 0.63 and a single other instance's 0.48, while the remainder alone stays at 0.04. By layer 16 the doc-last span holds something of the question that other instances' vectors lack and the remainder cannot supply on its own.

The geometry agrees and says why, before any injection: the cosine between two instances' d in this format is 0.998 at layer 0, 0.984 at layer 8 and 0.914 by layer 32, with the fraction of energy along the instance-mean falling from 0.997 to 0.880. At layer 8 the vector is 98% shared, so there is almost no instance-specific part to use; whether the remaining 12% at layer 32 ever matters is a question we can pose but have not answered.

Everything else replicates across the two prompt orders (pre-fix). Another skill's vector recovers 0.18, the mean over every skill 0.06, the same vectors with the span's positions permuted 0.00 and norm-matched noise 0.00; rescaling to the receiver's own norm costs nothing (0.99); and the first half of the span carries 1.05 against the second half's 0.43, the same concentration in the formula that the document-first format shows.

Where the document sits changes how deep the handover reaches. Placing the document after the question moves the cliff. We first reported this as a confirmed rank prediction, on the grounds that the doc-last span's participation ratio past layer 16 read 28–33 against the doc-first format's 14–17; §4.7 shows that both of those numbers are massive activations rather than representations, and once they are excluded the two formats have the same, flat dimensionality. The behavioural difference is real and the rank explanation for it is withdrawn.

The difference is large. On the same items, before the fix, as a fraction of each format's own range:

layer	0	4	8	12	**16**
document first	1.00	1.07	1.14	1.00	**0.14**
document last	0.99	1.11	1.00	0.93	**0.80**

Both formats carry the document's whole effect through layer 12. At layer 16 the document-first format carries a seventh of it and the document-last format four fifths; the latter's own cliff comes later, between layers 16 and 20 (0.82 to 0.35). Under causal masking a document placed first cannot see the question and a document placed last can, so the doc-last span's states are already question-conditioned when they are captured—which is the obvious candidate explanation, and is not one we have tested.

After the fix the layer-16 contrast holds: 0.85 with the document last ($n = 28$) against 0.19 with it first on forty calculators. Before the fix, at layers 0, 12, 16, 20 in the document-last format the mean of the skill's other instances tracked the instance's own (1.06/1.00,

0.88/0.94, 0.82/0.82, 0.29/0.35) with the remainder at the receiver; the rerun at layer 16 (0.63 against 0.85, above) does not reproduce that agreement, so we no longer claim that the instance-specific part never matters at depth—only that it does nothing alone.

What this rules out is the reading in which the handover depth is a fixed property of the network: it moves under a change to the prompt that leaves the model, the document and the items alone.

F.2 Depth, sufficiency and necessity, in full

On the clinical material the two questions—*can* the document's states be handed over at layer L, and *is* the model still reading them at layer L—give different answers, and the disagreement is the informative part.

Sufficiency, on forty calculators after the instrument fix ($n = 60$, twenty calculators after the restriction): a single layer's worth of the span reproduces $\rho = 1.04, 0.94, 0.85, 0.77$ at layers 0, 4, 8, 12 and then 0.28, 0.19, 0.19, 0.15 at 14, 15, 16, 20. The first, pre-fix sweep on twenty items (Table 11) has the same shape: 1.00 at 0 and 12, three quarters at 13–14, 0.14 at 15 and 16, nothing by 24.

Necessity, from cumulative attention knockout with the full chain of thought decoded ($n = 40$ rescued items, twenty calculators): masking every position's attention to the document's tokens from layer L to the end leaves 0.08–0.15 of the items correct for every $L \leq 20$ except $L = 4$ (0.33), then 0.38 at 24, 0.75 at 28 and 0.95 at 32. An earlier knockout run capped generation at 400 tokens, truncating the chain of thought so that even the unmasked prompt scored 0.50; it is superseded and not used. The model is still reading the document long after a single-layer handover has stopped working.

It is not that one layer is too few: the channel closes. The natural reading of the cliff is that a single layer stops being enough, since writing the span at *every* layer at once still recovers 0.950. It is wrong, and one experiment separates the two. We write the span at every layer in a window $[\ell_{\text{lo}}, \ell_{\text{hi}}]$ and nowhere else. The complete corrected window run uses a different receiver from the span battery. Using its own baselines and group screen retains 222 items from 21 calculators. Recovery is 0.95 for 8–11, 0.81 for 12–15, 0.22 for 14–19 and 0.10 for 16–35, with an exact identity control at 0.00. The final window matches layer 16 alone. Earlier versions incorrectly merged this run with the span battery's receiver baselines, producing $n = 137$ and incompatible normalisation; those estimates are superseded. Repeating the injection across late layers does not restore the early-window recovery, although the window and single-layer sweeps are not directly paired across receivers.

This is worth stating as a negative constraint on any mechanism, including the one we tried to give it. Whatever ends at layer 15 is not a capacity that more depth can compensate for.

The span's representation does not *become a summary, and the measurement that says it does is an artefact.* The obvious explanation for the cliff is that the span has by then been compressed, and the obvious measurement agrees: the centred participation ratio of the span's states, which counts the directions its positions differ in, reads 64 at layer 12 and 17 at layer 16—the same interval. We report it here because we believed it, built a mechanism on it, and it is wrong.

Three things give it away. It is a step and not a slope: the span states' participation ratio lies between 47 and 75 for every item at layer 14 and below 5 for thirty-four of forty at layer 16 (Table 13), and the content matrix's mean of 17 at layer 16 describes no item—thirty-two of forty read below 5 and the other eight at 81. A participation ratio of 1.0 over four hundred positions asserts that the centred span is rank one, which is not a statement one can make about four hundred tokens of text that the model still has to continue. And the collapse is absent in Qwen3-0.6B and Qwen3-1.7B at every depth (45–99, no item below 5) although their transplant cliff is in the same place in relative depth (§4.8).

The cause is measurable and known. At layer 16—and at no earlier layer—one token in 34 of 40 documents acquires a state whose norm is 99 to 180 times the span's median, carried by hidden dimensions 2276 and 676, the same two in all 34. That is the massive-activation phenomenon of Sun et al. [37], which appears at a single layer per model family [33]; in this model the loud token is a late delimiter inside the document rather than the sequence start. A centred spectrum is then owned by that one position: the top eight dimensions hold 97.7% of the variance at layer 16 against 5.4% at layer 14. Removing the four loudest positions out of four hundred restores the participation ratio to 66; unit-normalising each position restores it to 68; removing the top eight dimensions gives 27. Measured any of those three ways, the span's dimensionality is *flat* from layer 0 to layer 28 and only falls in the last few layers (Table 13).

So the honest statement is the narrower one. The single-layer handover dies between layers 14 and 16; the document's representation does not lose dimensionality there; we do not know what it does lose. What §4.6 establishes is a different and still causal fact—how many directions the injected object needs—and in the small pre-fix repeats at layers 12 and 16 that number is not lower either (Appendix G.1).

A warning about participation ratio on residual streams. Effective rank, participation ratio and intrinsic dimension are routinely computed on residual-stream activations, including in work that reports collapses with depth [11, 32]. Massive activations begin at a fixed layer, are carried by a handful of dimensions at a handful of positions, and are large enough to own a spectrum by themselves. Any such measurement that does not exclude them will report a collapse at exactly the emergence layer whether or not the representation has changed. Our own draft did. We recommend reporting the per-position norm ratio alongside any spectral statistic; here its median goes from 1.6 to 150 across the layer in question.

F.3 Where in the network the content is readable, in full

Transplanting one group of positions from the with-document forward into the without-document forward asks where the content can be *read from*; masking attention to the document asks where it is still being *read*. In the multiple-choice format the first question gives three windows that hand off cleanly (Figure 8, Appendix E.3): the document's own 688 tokens work at layers 0–10 and fall off a cliff at 11, the question's 42–47 tokens work at 11–16, and the final prompt position works at 21–27. The handoff is not an artefact of

Table 13: Centred participation ratio of the content matrix across the document's span on Qwen3-8B, measured four ways, medians over 96 items from all 48 documents (the plain column is bimodal, so its mean describes no item). The first column is the quantity usually reported and the only one that collapses. The last two columns say why—they are properties of the state itself, not of the difference: at layer 16 one token in 35 of 48 documents acquires a state 153 times (median) the span's norm, carried by two fixed hidden dimensions, and the top eight dimensions then hold 98% of the variance. Removing four positions out of four hundred, or normalising each position to unit length, removes the collapse entirely.

	participation ratio, computed…				massive activations	
layer	as usual	drop 4 loud	unit-norm.	drop 8 dims	max/med $\|h\|$	top-8 var.
8	97.7	97.0	103.4	114.2	1.5	4.9%
12	68.1	70.0	79.4	88.9	1.6	5.6%
14	76.3	76.8	83.7	93.7	1.6	5.3%
16	1.0	85.2	91.4	36.6	152.7	97.7%
20	1.1	84.9	89.1	70.1	94.2	94.8%
28	2.5	62.4	70.9	100.8	27.2	64.7%
34	32.4	34.2	43.0	113.4	1.3	15.7%

a shared analysis—the two experiments were run independently—and at layer 11 the document-span channel drops from 0.308 to 0.128 while the question-span channel rises from 0.231 to 0.308. The windows survive a change of model and not a change of format (Table 8): in free-form, on the same model and items, the question span recovers 0.000–0.025 at every one of 12 layers.

The second question gives a different shape. No single pathway is necessary: masking only the final position's attention to the document costs nothing (0.436 against 0.436) and masking only the question span's leaves 0.282, suggesting that these reader groups can partly substitute for one another. Removing access to the whole document has a larger effect in this setting. On third-party skills, making them unattendable for the whole forward takes accuracy from 0.550 [0.400, 0.700] to 0.075 [0.000, 0.175], against 0.025 [0.000, 0.075] with no document in the prompt at all—ninety per cent of the benefit, with the remainder inside the no-document interval. Masking an equal-length span of the patient note instead costs less, 0.150 [0.050, 0.275]. A genuinely inert length-matched control cannot be built here, since the only other long span is the data the task needs, so this bounds "removing 900 tokens" loosely but in the right direction.

Single-layer knockouts measure nothing on a 688-token span, because a layer denied the read recovers it one layer later; cumulative knockouts are required and are what we report. Cumulatively, the document stops being needed at layer 22 of 28 when the answer is one token, and is still being read at 83% depth when the answer is several hundred—the necessity-side view of §4.2.

F.4 What the content component belongs to, in full

If d were a "skill vector" it would be reusable: extracted from one task it would work on another. Its *direction* is reusable. Its *effect* is not, and the gap between those two statements is the finding.

The vector that carries contains no task information, and we check rather than argue it. Everything in this section so far is about the vector at the final position, the one that does not act. The vector that does act—the content component over the document's own span—has a property that makes the question sharp. In the prompt format the behavioural runs use, the document precedes the question, so causal masking makes the document's hidden states a function of the system text and the document alone. Within one skill the document is byte-identical across its instances, so d(span) should be *the same tensor* for every instance of that skill.

It is. Across the ten calculators in the causal set, the document span occupies token positions 43 to $43+m$ in every prompt, the token ids there are pairwise identical across instances (zero differing positions in 40 notes), and truncating two prompts to the end of the span and re-running the forward gives a relative difference of 0.0. In the full prompts the states differ by a relative 3.5×10^{-3} at layer 0 and 8.6×10^{-3} at layer 8, with zero positions below cosine 0.999—accumulated bf16 rounding from the differing sequence lengths, not content. The decode arms show the same thing behaviourally: the mean of the *other* instances' vectors and the instance's own agree to three decimals at every running mean, and the component of d_i orthogonal to that mean lands exactly on the receiver's baseline.

So the vector that reproduces the document's entire effect is a function of the document and of nothing else. Whatever a "skill vector" should be, this one is not a joint encoding of skill and task; it is the skill, and it is sufficient. The complementary experiment—moving the document behind the question so that its states *can* see the task—is §F.1.

The direction differs by skill; the effect does not transfer. On the clinical material each skill has 20 instances, so "same skill, different task" is directly measurable. Content vectors from different patient notes under the same calculator have cosine 0.789 at layer 20 and 0.876 at layer 30; vectors from different calculators, 0.320 and 0.653 ($n = 40$, Qwen3-8B). On the synthetic tier the analogous split is by conversion family, 0.869 within against 0.705 across at the last layer. The direction of d carries the identity of the document, increasingly so with depth. Its effect at the final position does not: injecting the mean direction $\bar{d}$, or another item's d_j, repairs 0.000 at every one of 16 layers in the free-form format against 0.268 for the item's own, and in multiple choice reaches 0.098, 0.293 and 0.049 for the mean, a same-family and a different-family donor against

0.780 for d_i (Table 26). The one band where a donor looks transferable is the band §4.2 identifies independently as one where the forward pass is damaged and every condition rises together.

Nothing about it distinguishes rescued items from persistent failures. The norms and angles of d and g agree to three digits across the two cells (layer 30, 8B, real skills: $\|d\|$ 189.4 vs 185.4, $\cos(d, g)$ −0.469 vs −0.468). A representational-similarity measure over the question span does separate them under the gold skill at layers 10–16 (−0.0037 to −0.0070, non-overlapping intervals, $n = 150 + 150$)—and separates them just as much under a length-matched *wrong* skill (−0.0035 to −0.0056). Zero layers pass a control the design required in advance, so what looked like a signature of "the skill worked here" is a signature of "a document is present": §E's trap one register down, and the reason we report a causal channel rather than a similarity score.

The content component's size depends on where it is written. At the final position the content component is a 6% perturbation of the state it displaces at layer 6 ($\|d\| = 2.3$ against $\|h_{\text{no}}\| = 37.3$), rising to 51% by layer 35—and it is causally inert at every one of them. Over the document's own span the same difference of the same two conditions is 95% to 115% of the state at every depth (Table 12)—a replacement rather than a nudge—and it is causally complete (§4.3). What distinguishes the two places is not how much the document is written into them; it is whether what is written is the whole of what the downstream computation reads.

The shared part carries the norm, not the effect. Between 73% and 99.7% of each item's d_i lies along the shared mean direction, depending on layer (Table 25); the remainder occupies an effective 3–18 dimensions out of 2048–4096. The reusable part is almost all of the norm and, in free-form answering, none of the effect; in multiple choice it is worth 0.06 of the 0.79 the item's own vector achieves. A single vector per skill is therefore a fair description of what the injection *looks like* and a wrong description of what it *does*, which is the same inversion as §E one level down.

G RANK TRUNCATION, THE MODEL LADDER AND REPLICATION

G.1 Rank truncation and the model ladder, in more detail

On the denser grid every rank curve in the paper reaches its untruncated value: the three TheoremQA rows at $k = 192$–384 (0.806/0.882/0.871 against 0.839 for Qwen3-8B, 0.717/0.701/0.732 against 0.740 for Qwen3-0.6B, and 0.819/0.880/0.819 against 0.940 for Mistral-7B at $k = 192, 256, 384$), MedCalc with Qwen3-8B and Qwen3-0.6B at $k = 256$, and MedCalc with Mistral-7B at $k = 384$ (0.842, exactly its untruncated value; 0.825 at 512). The knee k^* is computed on the fixed grid $\{16, 32, 64, 128\}$ and is unchanged by the added points, by construction.

Every measurement so far says *where* the document's effect lives. This one says *how big it is*, in the only unit that is comparable across the task-vector literature: the number of directions of the injected object that the model actually uses.

We truncate the content matrix across the span's positions to rank k before writing it. With $d \in \mathbb{R}^{m \times D}$ and $\bar{d}$ its mean over positions,

$$d^{(k)} = \bar{d} + \sum_{i=1}^{k} \sigma_i u_i v_i^\top, \qquad d - \bar{d} = \textstyle\sum_i \sigma_i u_i v_i^\top, \tag{5}$$

the mean kept and only the variation truncated, so that k is comparable with a centred participation ratio. $k = 1$ is the closest thing in this setting to a task vector: one direction, applied at every position of the span.

One direction is not enough, and neither are sixteen (Figure 3). On the full rerun's screened subset after the fix ($n = 135$, 14 calculators), recovery at layer 8 is 0.07, 0.05, 0.12, 0.63, 0.77 at $k = 4, 16, 32, 64, 128$ against 0.89 untruncated, and half of the curve's range above $k = 16$ is recovered at $k^* = 47$ (95% bootstrap interval over calculators $[43, 55]$; interpolated in $\log_2 k$, the definition fixed before the model ladder ran). The ten-calculator development curve, before the fix, reads 0.11, 0.06, 0.06, 0.11, 0.11, 0.33, 0.72, 0.94 at $k = 1, 2, 4, 8, 16, 32, 64, 128$ and gives $k^* = 45$.

k^* is anchored to the largest value on the grid $k \le 128$, not to the untruncated transplant. Where $k = 128$ is close to untruncated (Qwen3-8B, 0.77 against 0.89) that makes little difference; where it is not (Mistral-7B, 0.71 against 0.92) the knee is bounded by the grid, and a finer or longer grid could move it up.

It is the top of the spectrum, not the count. Keeping the *bottom* k singular directions instead—the same rank, the discarded half of the spectrum—gives $\rho = 0.06$ at $k = 16$ and 0.00 at $k = 64$ on forty calculators (0.00 and 0.06 on the development set), both at the receiver. Without that control, "sixty-four directions are enough" could not be distinguished from "any sixty-four directions will do".

The requirement does not fall with depth (pre-fix, small n). If the handover died at layer 15 because the object had shrunk, the number of directions needed would shrink with it. Repeating the truncation where the transplant still works, on the calculators the receiver never solves, gives accuracy 0.00 at $k = 16$ and 0.62 at $k = 128$ against 1.00 untruncated at layer 12 ($n = 16$), and 0.25 at $k = 16$ and 0.71 at $k = 128$ against 0.79 at layer 16 with the document after the question ($n = 28$)—the knee between 64 and 128 in both. These runs predate the fix and are small; we read them as "not lower", not as a measured knee. At layer 16 with the document *before* the question, where the transplant is already at the floor ($\rho = 0.06$), there is no curve at all: every truncation reads between 0.00 and 0.06, because there is no longer an effect for truncation to remove. We do not read the rank of the layer-16 doc-first object at all, since that is the layer at which the massive activations of §4.7 take over its spectrum.

The obvious converse does not work, and we report that. If a deep layer's handover failed for want of directions, supplying it with a higher-rank message should restore it. We tried the direct version—write the layer-8 donor into layer 16—on ten calculators ($n = 30$ after restricting to items that neither the wrong document nor the no-document prompt solves). It does not restore; it reads $\rho = 0.00$ where the same-layer donor reads 0.06, and matching the norm to layer 16's scale recovers only part of the gap (0.13). The layer-8 state is not a high-rank version of the layer-16 state, it is a different

Table 14: The channel and the truncation curve at three model widths, each with its own restricted split and its own injection layer at matched relative depth. ρ is Eq. 3; *gold* is what the document itself is worth on that model's restricted items, which is why the 0.6B and 1.7B rows have low absolute accuracy and a large n: those models solve almost nothing without the document, so almost every item qualifies. Qwen3-0.6B and Qwen3-8B are after the instrument fix; Qwen3-1.7B is before it, and its curve is not monotone, so its k^* is not interpretable. k^* sits between 32 and 64 at $d = 1024$ and at $d = 4096$ alike; a width-proportional account predicts $k \approx 12$ for the first. Mistral-7B is in Table 15.

model	d	layers	inject	n	gold	ρ(full)	k=16	32	64	128	bottom-64	k^*
Qwen3-0.6B	1024	28	6	204	0.260	+0.83	+0.02	+0.23	+0.81	+0.85	+0.02	41
Qwen3-1.7B	2048	28	6	267	0.090	+0.54	+0.33	+0.50	+0.42	+0.29	+0.12	23
Qwen3-8B	4096	36	8	135	0.919	+0.89	+0.05	+0.12	+0.63	+0.77	+0.03	47

code, and the transfer is directional: Ardoin et al. [2] find middle-to-early injection optimal, and we find early-to-middle injection harmful.

G.2 Replication across tasks and model families, in full

Every point behind Table 4. All runs are after the instrument fix. MedCalc rows use forty calculators and each model's own behavioural screen; TheoremQA rows use SRA-Bench's 747 questions with its prompt and scorer verbatim, and since most of its 320 skills serve a single question the receiver screen is per skill group, and reduces to an item-level screen for singleton skills. Those 747 questions are the pool each TheoremQA run samples from, not the set it evaluates: a run takes at most four questions per skill and at most two hundred skills, which here is 142–168 items over 113–122 skills before the screen. The MedCalc rows are not sampled this way, and the two should not be read as equally exhaustive. Knockout rows are fractions of each model's rescued items kept when attention to the document is blocked from layer L to the end, unrestricted.

Three observations are not in the main text. (i) On TheoremQA with Qwen3-8B, blocking attention to the document from layer 0 still keeps 0.31 of the rescued items, against 0.12 on MedCalc. Blocking attention is not removing the document—its positions still shift the question's rotary phase and length —and the receiver restriction selects failures near the model's decision boundary; we have not separated the two. (ii) The same receiver-based selection is the most likely reason TheoremQA's controls sit at 0.19–0.25 for Qwen3-8B. (iii) On LogicBench (not shown), the method's resolution fails: with mostly binary answers the restricted split ($n = 57$, 16 skills) gives the transplant 0.75 $[0.60, 0.88]$, its own d on the shared prefix 0.80, half a dose 0.44, another skill 0.38, permuted 0.25 and noise 0.15, and the depth curve has no cliff (0.93 at layer 0, 0.49 at 16, 0.33 at 20).

H RELATED WORK, AT LENGTH

Skill benchmarks measure the effect, not the mechanism. A 2026 literature evaluates skill documents behaviourally: across domains [22], on executable software-engineering tasks [14], and under different delivery protocols [24]; a parallel line studies which skill to retrieve. Closest in motivation is distilling a skill into weights [47], which asks what a skill *could* become rather than what it already is at inference time. None of this work looks inside the forward pass, and the effect sizes it reports are exactly what our §4.2 shows can be format-dependent.

Task and function vectors, and the two limits on what one carries. That a context can be compressed into a single activation-space direction is established for in-context demonstrations [17, 40] and, in weight space, for fine-tuning [18]. Methodologically we are in this line: our d is extracted and re-injected the same way. Two limits on that move are now known, and they are not the same limit.

Dong et al. [9] prove the first. In a linear transformer, injecting one task vector is functionally a rank-one map, so a task whose input–output relation is high-rank—a bijection, for instance—cannot be encoded by a single vector; they confirm it on Llama-7B/13B and repair it by injecting several vectors at several demonstration positions. That result *explains one of ours*: on the synthetic tier, $\bar{d}$—one vector shared by the whole item set, which is what a reusable skill vector would be—recovers 0.098, and the mapping it would have to encode is a 39-row lookup table. We do not claim that null as new; we attribute it.

The limit this paper is about is different, and the two can be separated empirically. On the same synthetic tier, the *item's own* d_i recovers 0.805 through the same single position despite that same high-rank mapping, because the answer is one token; and on clinical skills the same injection recovers nothing because the answer is hundreds of tokens. Rank constrains which mappings one vector can express; length constrains how much of a decision any small position group can carry, whatever the mapping. Dura et al. [10] state the length limit as a motivation for patching CoT sequentially rather than at a static position; §4.2 turns it into a controlled measurement, and §4.6 measures the rank of the *representation* rather than of the mapping.

We also differ in position. Function and task vectors are read from, and written to, one position, and are validated on tasks whose answer is a single token. Mehrafarin et al. [27] show how far that can be pushed and, read against our result, show what the position holds: patching a single late position with a state taken from the model's *own completed* chain of thought recovers the correct GSM8K answer at least 99% of the time, while our donor—taken from the prompt, before any reasoning—recovers nothing. Same position, same intervention; what differs is whether the answer already exists in the donor. Da Silva et al. [7] report a reliability failure of the same channel from another direction: across 36 models, function vectors with default parameters recover 5-shot performance in 20% of model–task pairs.

Table 15: The replication curves. Battery, sufficiency and rank are ρ (Eq. 3) on the restricted split; necessity is the fraction of rescued items kept. "Another family" (MedCalc) and "another skill" (TheoremQA) are the transfer controls available in each set.

task / model	measurement	values
MedCalc / Qwen3-8B	battery, L8 (n = 135, 14 groups)	transplant 0.89, own d on shared prefix 0.65, α=0.5 0.04, another skill 0.05, another family 0.05, same family 0.12 (n = 41), permuted 0.05, noise 0.02, null 0.00
	sufficiency (n = 135)	L0 0.98, L1 0.95, L2 0.94, L3 0.96, L4 0.98, L5 0.94, L6 0.94, L7 0.92, L8 0.89, L9 0.85, L10 0.89, L11 0.81, L12 0.74, L13 0.16, L14 0.16, L15 0.06, L16 0.06, L17 0.07, L18 0.03, L19 0.02, L20 0.03, L21 0.02, L22 0.04, L23 0.03, L24 0.01, L25 0.02, L26 0.03, L27 0.03, L28 0.02, L29 0.03, L30 0.05, L31 0.04, L32 0.04, L33 0.02, L34 0.01, L35 0.00
	necessity (n = 134 rescued, screened groups)	L0 0.03, L1 0.01, L2 0.03, L3 0.01, L4 0.04, L5 0.05, L6 0.03, L7 0.03, L8 0.04, L9 0.03, L10 0.03, L11 0.04, L12 0.04, L13 0.07, L14 0.08, L15 0.07, L16 0.07, L17 0.08, L18 0.06, L19 0.07, L20 0.12, L21 0.08, L22 0.09, L23 0.08, L24 0.25, L25 0.58, L26 0.46, L27 0.56, L28 0.57, L29 0.57, L30 0.78, L31 0.84, L32 0.85, L33 0.88, L34 0.91, L35 0.94
	rank, L8 (n = 135)	k=4 0.07, k=16 0.05, k=32 0.12, k=64 0.63, k=128 0.77, full 0.89, bottom-16 0.04, bottom-64 0.03, k^* 47
TheoremQA / Qwen3-8B	battery, L8 (n = 99, 75 groups)	transplant 0.84, own d on shared prefix 0.86, α=0.5 0.19, another skill 0.25, permuted 0.24, noise 0.22, null 0.00
	sufficiency (n = 99)	L0 1.00, L1 1.01, L2 0.98, L3 0.94, L4 0.94, L5 0.96, L6 0.95, L7 0.91, L8 0.84, L9 0.75, L10 0.82, L11 0.67, L12 0.61, L13 0.44, L14 0.37, L15 0.22, L16 0.19, L17 0.17, L18 0.19, L19 0.19, L20 0.14, L21 0.13, L22 0.09, L23 0.11, L24 0.14, L25 0.12, L26 0.16, L27 0.10, L28 0.16, L29 0.12, L30 0.09, L31 0.14, L32 0.04, L33 0.09, L34 0.08, L35 0.00
	necessity (n = 63 rescued, screened groups)	L0 0.25, L1 0.25, L2 0.13, L3 0.19, L4 0.24, L5 0.21, L6 0.24, L7 0.25, L8 0.27, L9 0.24, L10 0.21, L11 0.24, L12 0.21, L13 0.29, L14 0.27, L15 0.19, L16 0.27, L17 0.30, L18 0.35, L19 0.43, L20 0.49, L21 0.57, L22 0.52, L23 0.63, L24 0.70, L25 0.81, L26 0.78, L27 0.81, L28 0.78, L29 0.84, L30 0.86, L31 0.90, L32 0.89, L33 0.92, L34 0.90, L35 0.98
	rank, L8 (n = 99)	k=4 0.17, k=16 0.24, k=32 0.30, k=64 0.54, k=128 0.67, full 0.84, bottom-16 0.10, bottom-64 0.13, k^* 50
MedCalc / Qwen3-0.6B	battery, L6 (n = 362, 38 groups)	transplant 0.91, own d on shared prefix 0.90, α=0.5 0.07, another skill 0.30, another family 0.04, same family 0.11 (n = 204), permuted 0.02, noise 0.03, null 0.00
	sufficiency (n = 362)	L0 1.05, L1 1.08, L2 1.03, L3 0.97, L4 1.00, L5 1.07, L6 0.91, L7 0.91, L8 0.96, L9 0.70, L10 0.67, L11 0.23, L12 0.10, L13 0.14, L14 0.12, L15 0.18, L16 0.13, L17 0.10, L18 0.13, L19 0.12, L20 0.12, L21 0.07, L22 0.02, L23 0.05, L24 0.07, L25 0.03, L26 0.01, L27 0.00
	necessity (n = 90 rescued, screened groups)	L0 0.03, L1 0.06, L2 0.03, L3 0.06, L4 0.04, L5 0.02, L6 0.06, L7 0.02, L8 0.06, L9 0.01, L10 0.03, L11 0.03, L12 0.00, L13 0.00, L14 0.02, L15 0.01, L16 0.02, L17 0.06, L18 0.04, L19 0.03, L20 0.03, L21 0.46, L22 0.76, L23 0.77, L24 0.84, L25 0.87, L26 0.89, L27 1.00
	rank, L6 (n = 362)	k=4 0.01, k=16 0.00, k=32 0.12, k=64 0.79, k=128 0.67, full 0.91, bottom-16 0.02, bottom-64 0.02, k^* 43
TheoremQA / Qwen3-0.6B	battery, L6 (n = 132, 97 groups)	transplant 0.74, own d on shared prefix 0.57, α=0.5 0.13, another skill 0.08, permuted 0.09, noise 0.10, null 0.00
	sufficiency (n = 132)	L0 0.97, L1 0.93, L2 0.91, L3 0.88, L4 0.88, L5 0.89, L6 0.74, L7 0.74, L8 0.69, L9 0.65, L10 0.64, L11 0.40, L12 0.33, L13 0.33, L14 0.31, L15 0.29, L16 0.28, L17 0.19, L18 0.22, L19 0.14, L20 0.13, L21 0.05, L22 0.06, L23 0.06, L24 0.03, L25 0.01, L26 0.00, L27 0.00
	necessity (n = 125 rescued, screened groups)	L0 0.11, L1 0.13, L2 0.10, L3 0.10, L4 0.10, L5 0.06, L6 0.09, L7 0.15, L8 0.17, L9 0.14, L10 0.14, L11 0.11, L12 0.20, L13 0.18, L14 0.18, L15 0.21, L16 0.27, L17 0.25, L18 0.34, L19 0.33, L20 0.54, L21 0.66, L22 0.86, L23 0.88, L24 0.90, L25 0.96, L26 0.97, L27 0.99
	rank, L6 (n = 132)	k=4 0.06, k=16 0.09, k=32 0.14, k=64 0.43, k=128 0.61, full 0.74, bottom-16 0.06, bottom-64 0.08, k^* 52
MedCalc / Mistral-7B	battery, L4 (n = 92, 20 groups)	transplant 0.84, own d on shared prefix 0.53, α=0.5 0.19, another skill 0.16, another family 0.28, same family 0.35 (n = 31), permuted 0.16, noise 0.14, null 0.00
	sufficiency (n = 92)	L0 0.96, L1 0.96, L2 0.89, L3 0.79, L4 0.84, L5 0.79, L6 0.75, L7 0.46, L8 0.46, L9 0.26, L10 0.21, L11 0.21, L12 0.14, L13 0.19, L14 0.23, L15 0.21, L16 0.09, L17 0.14, L18 0.25, L19 0.07, L20 0.09, L21 0.12, L22 0.09, L23 0.04, L24 0.11, L25 0.14, L26 0.11, L27 0.11, L28 0.12, L29 0.18, L30 0.09, L31 0.00
	necessity (n = 46 rescued, screened groups)	L0 0.04, L1 0.09, L2 0.11, L3 0.20, L4 0.15, L5 0.20, L6 0.15, L7 0.15, L8 0.09, L9 0.09, L10 0.15, L11 0.09, L12 0.33, L13 0.22, L14 0.24, L15 0.24, L16 0.26, L17 0.28, L18 0.28, L19 0.28, L20 0.17, L21 0.28, L22 0.46, L23 0.41, L24 0.41, L25 0.41, L26 0.43, L27 0.50, L28 0.65, L29 0.70, L30 0.70, L31 0.85
	rank, L4 (n = 92)	k=4 0.16, k=16 0.14, k=32 0.09, k=64 0.30, k=128 0.54, full 0.84, bottom-16 0.11, bottom-64 0.14, k^* 72
TheoremQA / Mistral-7B	battery, L4 (n = 115, 94 groups)	transplant 0.94, own d on shared prefix 0.80, α=0.5 0.27, another skill 0.20, permuted 0.17, noise 0.13, null 0.00
	sufficiency (n = 115)	L0 1.01, L1 1.05, L2 0.96, L3 0.92, L4 0.94, L5 1.00, L6 0.82, L7 0.49, L8 0.42, L9 0.40, L10 0.30, L11 0.19, L12 0.11, L13 0.12, L14 0.11, L15 0.08, L16 0.06, L17 0.10, L18 0.06, L19 0.06, L20 0.05, L21 0.07, L22 0.06, L23 0.08, L24 0.06, L25 0.07, L26 0.06, L27 0.07, L28 0.10, L29 0.05, L30 0.07, L31 0.00
	necessity (n = 65 rescued, screened groups)	L0 0.09, L1 0.05, L2 0.14, L3 0.08, L4 0.06, L5 0.05, L6 0.05, L7 0.02, L8 0.05, L9 0.05, L10 0.12, L11 0.11, L12 0.18, L13 0.34, L14 0.31, L15 0.37, L16 0.40, L17 0.40, L18 0.43, L19 0.60, L20 0.66, L21 0.80, L22 0.88, L23 0.82, L24 0.85, L25 0.89, L26 0.89, L27 0.88, L28 0.89, L29 0.91, L30 0.86, L31 0.94
	rank, L4 (n = 115)	k=4 0.17, k=16 0.20, k=32 0.23, k=64 0.48, k=128 0.76, full 0.94, bottom-16 0.14, bottom-64 0.13, k^* 64

When one vector is *enough, and what that tells us.* The strongest recent positive result for single-vector compression is Ardoin et al. [2], who learn a weighted sum of an instruction prompt's activations at a middle layer, inject the resulting *single* vector at one placeholder position in an early layer, and lose under two points of accuracy. We read it as convergent rather than contradictory, for two reasons, and both are load-bearing here. First, every task on which it succeeds has a short answer: capitals, ISO codes, continents, years, word translation, antonyms, currency codes—all single-word exact match—and ARC-Easy, which is a single letter. That is exactly the half of the answer-length axis where our own single-position patch also works (0.805 on the synthetic tier, §4.2); the half where it fails is the one their evaluation does not enter. Second, their optimal configuration extracts at a middle layer and injects at an *early* one—their stated reason is that an early injection leaves more layers to integrate it—which is independently the direction of our depth result (§4.7). Taken with Dong et al. [9] and Wiegreffe et al. [44], the three give one sentence: a single vector is a *selector* among behaviours the model already has—their appendix encodes nine tasks in one vector, about three bits—and not a carrier of content that needs a hundred directions to rebuild.

Three different objects are called "rank" here. Because this paper adds a fourth, we separate them explicitly. Dong et al. [9] bound the rank of the *task map*: which input–output relations one injected vector can express. Sarkar and Deka [32] measure the rank of the *Jacobian*, eff-rank $\mathbb{E}_x[J_L^\top J_L]$—how many directions of a layer causally reach the output—and report it to be layer-intrinsic and, notably, unchanged between Gemma-2-2B and Gemma-2-9B. Fernando and Guitchounts [11] measure the rank of the *composed operator* across depth and find it collapses by about 65× in Llama-3.1-8B. We measure the rank of a *representation*: the matrix $d \in \mathbb{R}^{m\times d}$ that one document contributes across its own m span positions. The distinction matters for what our result means. An earlier draft combined a layer-intrinsic causal capacity [32] with the participation ratio of d falling from 64 to 17 to argue that the handover closes on the *message* side; that fall is massive activations (§4.7), and with them excluded the ratio is flat, so we withdraw the argument and do not know on which side the window closes. We are also, among these four, the only one to truncate the injected object itself and read out behaviour; the other three use a theorem, a first-order attribution proxy, and a spectral surgery on the weights.

A caution applies to all of the descriptive ones, ours included. Sun et al. [37] document activations three orders of magnitude above the median, carried by a fixed pair of hidden dimensions at a small set of token positions, appearing at a single layer [33] and acting as bias terms. A spectral statistic computed on a residual stream past that layer is a statistic about those activations. Our own centred participation ratio reads 1.0 over four hundred positions at layer 16 of Qwen3-8B and 66 once four positions are dropped (§4.7); an uncentred entropy rank, which is what Fernando and Guitchounts [11] report collapsing by 65×, is if anything more exposed, since the massive component is not removed by centring there either. We do not claim their result is an artefact—their object is the composed operator, not a representation—but the check is cheap and we think it should be standard.

Compressing a document into parameters or into few tokens. A parallel line replaces the document with weights rather than activations— hypernetworks that emit LoRA parameters from a document [6], and context distillation into a knowledge module in both output and hidden space [5]. These are orthogonal to us in method (weights, and training, versus activations and none) but share the premise that a document's effect is compressible. Wang et al. [43] report that re-supplying the context after such internalisation *degrades* accuracy, which is a phenomenon our $t = g + d$ split gives a vocabulary for: what is internalised is not the same object as what the document contributes while present.

Activation patching. Activation patching tests recovery in a specified receiver, while knockouts test dependence under a specified removal intervention [16, 46]. Pola and Balasubramanian [31] compare instruction-span and final-position patches between minimal-contrast prompts. Our main-text depth comparison pairs span replacement with cumulative attention knockout on matched screened items. Li et al. [23] distinguish transferable states from decodable task identity and find temporal and semantic locality. This motivates comparing positions, while continued reading requires a separate intervention. Instruction-based function vectors can use different components from example-based ones [8], and patches at the final instruction token can interact across layers [4]. Abdelsalam et al. [1] likewise localise recoverable causal signal to a changed cue region plus a few intermediate tokens before aggregation near the output. Mechanistic work on retrieval-augmented generation [42] patches the residual stream to localise attribution but does not separate presence from content.

I ADDITIONAL TABLES

I.1 Additional diagnostics for the main-text argument

On the synthetic Qwen3-1.7B task (358 items), the effect extends beyond rescued items: across rescued, persistent-failure, kept and broken items, replacement gives accuracies of 1.000/0.000/1.000/0.000, while content injection gives 0.805/0.075/1.000/0.186. Content injection thus broadly reproduces both the benefits and the errors induced by the skill.

Readout checks. Alternative readouts can rank interventions differently. On the synthetic task, averaged gold-answer log-probability favours a mean shared across skills by 5.05 nats despite lower accuracy (0.251 versus 0.288, 358 items). Forcing an `ANSWER:` cue on MedCalc cuts the skill's accuracy gain from 39.7 to 4.7 points (0.178 versus 0.131 on 720 items) while leaving a +1.08 nat log-probability difference. Synthetic multiple choice retains 0.54 of the skill's effect after its numbers are corrupted and 0.82 after its lines are scrambled, illustrating possible answer-format shortcuts [3] (Appendix D). Other domains give limited evidence (Appendix G.2).

The receiver screen affects how these controls should be interpreted. On the 34 excluded calculators, the wrong-skill receiver already solves some items. Other-skill content recovers 0.17 on these groups ($n = 332$), compared with 0.05 on the retained groups ($n = 135$). Noise recovers 0.10 versus 0.02. This pattern is consistent with perturbations enabling knowledge already available to

Table 16: Where the aggregate log-probability gap comes from, synthetic control, layer 21, $n = 39$. A cell's contribution to the mean is its size share times its own gap, so the column adds up exactly. The mean vector beats the true state overall because F and B—the cells where the document does not work—contribute 133% of the gap; on R, where it does work, the true state wins by 2.5 nats and by 0.69 in accuracy.

cell	n	mean − real (nats)	contribution to +2.747	acc. real	acc. mean
R rescued	13	−2.506	−0.835 (−30%)	**0.923**	0.231
F persistent	17	+5.907	+2.575 (+94%)	0.059	0.294
K kept	4	−0.740	−0.076 (−3%)	1.000	0.500
B broken	5	+8.455	+1.084 (+40%)	0.000	0.200

Table 17: The numbers behind Figure 1, restricted as the figure is to the calculators whose receiver never solves an item.

	accuracy	CI_{95}
the gold document in the prompt, and no document at all		
gold document in context	0.950	[0.854, 1.000]
no document	0.050	[0.000, 0.146]
written into the last prompt position (best layer of 10, 14, 20, 30 for each arm)		
replace the state with h_{skill}	0.150	[0.000, 0.306]
$h + d$	0.150	[0.000, 0.306]
$h + g$	0.050	[0.000, 0.146]
$h + d_j$, another instance of this skill	0.150	[0.000, 0.306]
$h + d_j$, another skill	0.100	[0.000, 0.231]
written over the document's own span, layer 8		
the filler document, unpatched (the floor here)	0.000	[0.000, 0.000]
$h_{\text{recv}} + d$ (= h_{skill}, $\alpha = 1$)	0.950	[0.854, 1.000]
$h_{\text{recv}} + 2d$	0.800	[0.625, 0.975]
$h_{\text{recv}} + \frac{1}{2}d$	0.050	[0.000, 0.146]
$h_{\text{recv}} + \tilde{d}$, shared prefix only	0.900	[0.769, 1.000]
$h_{\text{recv}} + \tilde{d}_j$, another skill, same prefix	0.050	[0.000, 0.146]
$h_{\text{recv}} + d$, span positions permuted	0.100	[0.000, 0.231]
h_{recv}+ noise, $\|\cdot\|$ matched per position	0.000	[0.000, 0.000]
h_{recv} itself (must be a no-op)	0.000	[0.000, 0.000]

Table 18: The document-span depth sweep over all forty items, unrestricted. Compare Table 11: the profile is the same shape with a higher floor, because the calculators this adds are ones a wrong document already rescues.

layer	accuracy	CI_{95}	fraction of the document's own effect
0	0.950	[0.882, 1.000]	+1.00 (no-op arm 0.150)
4	0.950	[0.882, 1.000]	+1.00
8	1.000	[1.000, 1.000]	+1.06
12	0.875	[0.773, 0.977]	+0.90 (no-op arm 0.150)
13	0.675	[0.530, 0.820]	+0.65 (no-op arm 0.200)
14	0.700	[0.558, 0.842]	+0.68 (no-op arm 0.175)
15	0.400	[0.248, 0.552]	+0.29 (no-op arm 0.150)
16	0.400	[0.248, 0.552]	+0.29 (no-op arm 0.175)
18	0.250	[0.116, 0.384]	+0.10 (no-op arm 0.175)
20	0.475	[0.320, 0.630]	+0.39
24	0.175	[0.057, 0.293]	+0.00
30	0.150	[0.039, 0.261]	−0.03 (no-op arm 0.150)
every layer	0.950	[0.882, 1.000]	+1.00
the filler document, unpatched	0.175		0.00
the gold document in context	0.950		1.00

Table 19: The same depth sweep run a second time against the other receiver—a single fixed wrong skill rather than each skill's own paired control—on the same items. The two agree wherever the first is well behaved; the one bump in the first run does not replicate.

layer	accuracy	CI_{95}	fraction of the document's own effect
0	0.950	[0.882, 1.000]	+1.00 (no-op arm 0.125)
4	1.000	[1.000, 1.000]	+1.06
8	0.975	[0.927, 1.000]	+1.03 (no-op arm 0.125)
12	0.800	[0.676, 0.924]	+0.81
16	0.375	[0.225, 0.525]	+0.28
20	0.275	[0.137, 0.413]	+0.16 (no-op arm 0.150)
the filler document, unpatched	0.150		0.00
the gold document in context	0.950		1.00

Table 20: The same arms over all forty items, unrestricted, after the instrument fix. The six calculators this adds are ones the model can partly do without the gold document: their receiver scores 0.375**, another skill's vector** 0.667 **and norm-matched noise** 0.375**, against** 0.000, 0.188 **and** 0.000 **on the four that need the document. Pooling them raises the transfer arm from** $\rho = 0.19$ **to** 0.32**, which is why the main text reports the restricted set.**

	accuracy	CI_{95}
the gold document in the prompt, and no document at all		
gold document in context	0.950	[0.854, 1.000]
no document	0.050	[0.000, 0.146]
written into the last prompt position (best layer of 10, 14, 20, 30 for each arm)		
replace the state with h_{skill}	0.150	[0.000, 0.306]
$h + d$	0.150	[0.000, 0.306]
$h + g$	0.050	[0.000, 0.146]
$h + d_j$, another instance of this skill	0.150	[0.000, 0.306]
$h + d_j$, another skill	0.100	[0.000, 0.231]
written over the document's own span, layer 8		
the filler document, unpatched (the floor here)	0.150	[0.039, 0.261]
$h_{\text{recv}} + d$ $(= h_{\text{skill}}, \alpha = 1)$	0.975	[0.927, 1.000]
$h_{\text{recv}} + 2d$	0.850	[0.739, 0.961]
$h_{\text{recv}} + \frac{1}{2}d$	0.050	[0.000, 0.118]
$h_{\text{recv}} + \bar{d}$, shared prefix only	0.950	[0.882, 1.000]
$h_{\text{recv}} + \bar{d}_j$, another skill, same prefix	0.425	[0.272, 0.578]
$h_{\text{recv}} + d$, span positions permuted	0.300	[0.158, 0.442]
h_{recv}+ noise, $\|\cdot\|$ matched per position	0.200	[0.076, 0.324]
h_{recv} itself (must be a no-op)	0.125	[0.023, 0.227]

Table 21: What the content vector contains once it is able to contain the task—the numbers behind Figure 10.

donor written over the document's span	accuracy	CI_{95}	fraction
the filler document, unpatched	0.000		0.00
the gold document in context	0.850		1.00
d_i, this instance's own	0.850	[0.694, 1.000]	+1.00
$\bar{d}_{\neg i}$, the mean over the skill's other instances	0.800	[0.625, 0.975]	+0.94
d_j, one other instance	0.900	[0.769, 1.000]	+1.06
$d_i^{\parallel}$, the half along that mean	0.850	[0.694, 1.000]	+1.00
$d_i^{\perp}$, the rest	0.000	[0.000, 0.000]	+0.00
$\bar{d}$, another skill	0.150	[0.000, 0.306]	+0.18
$\bar{d}$, averaged over every skill	0.050	[0.000, 0.146]	+0.06
the same d_i, span positions permuted	0.000	[0.000, 0.000]	+0.00
noise, norm matched per position	0.000	[0.000, 0.000]	+0.00

Table 22: Three transplant channels, synthetic control, Qwen3-1.7B, $n = 39$. Each row is a separate experiment; the receiver baseline differs because the document-span and question-span transplants run inside a prompt that still contains a filler document.

channel	positions transplanted	effective layers	peak acc.	receiver baseline
document span	the document's own 688 tokens	0–10	0.436 (L3–4)	0.154
question span	the question's 42–47 tokens	11–16	0.333 (L12/14)	0.154
final position	the last prompt token	21–27	0.436–0.487	0.231

Table 23: The same measurements at two model sizes, on the 358-item reruns. The effect itself, in both formats, is the top block of Table 7. The multiple-choice presence row is the reason we take the free-form format as the primary evidence: on the 8B, chance on four options is 0.250 and g sits at 0.298, within a few points of it, while in free-form it reads 0.008.

	Qwen3-1.7B	Qwen3-8B
MC, rescued cell: replace outright	1.000	1.000
MC, rescued cell: $+d$	0.805	0.894
MC, rescued cell: $+g$	0.098	0.298
free-form, rescued cell: replace outright	—	0.235
free-form, rescued cell: $+d$	—	0.210
free-form, rescued cell: $+g$	—	0.008
layers where $\|g\| > \|t\|$	0–16	0–12
PR(d) centred, deepest layer	5.4	18.1
$\cos(d_i, d_j)$ same / other family, deepest	0.869 / 0.705	0.584 / 0.530

Table 24: The content component split along its own shared direction and orthogonal to it, rescued cell, $n_R = 82$. The crossover at layer 21 is the same layer at which cumulative attention knockout on the document becomes free (§4.7) and at which another item's content vector stops working (§4.5).

injected	L16	L18	L20	L21	L24	L26
$h_{\text{no}} + d_i$ (whole)	0.134	0.390	0.439	0.500	0.695	0.805
$h_{\text{no}} + (d_i \cdot u)u$ (shared dir.)	0.122	0.305	0.268	0.098	0.049	0.085
$h_{\text{no}} + d_i^{\perp}$ (item-specific)	0.012	0.122	0.195	0.427	0.524	0.561
$h_{\text{no}} + \bar{d}$ (shared dir., fixed norm)	0.134	0.305	0.293	0.122	0.049	0.085

Table 25: The decomposition of Eq. 1, synthetic control, Qwen3-1.7B, fp32, $n = 358$, at the last prompt position. PR is the participation ratio of $\{d_i\}$ after removing their shared mean direction: the effective number of dimensions the item-to-item variation occupies, out of 2048. The last column is the fraction of a single item's d_i that lies along that shared direction.

layer	$\|d\|$	$\|g\|$	$\|d\|/\|t\|$	$\cos(d, g)$	$\cos(d_i, d_j)$		PR(d)	on mean
					same fam.	cross fam.		
0	1.01	2.66	0.43	−0.50	0.996	0.995	3.2	0.998
10	16.24	38.96	0.45	−0.38	0.989	0.987	4.5	0.994
14	33.47	68.53	0.52	−0.37	0.979	0.970	8.6	0.987
21	459.14	352.60	1.04	−0.43	0.920	0.794	5.5	0.919
27	919.82	704.45	1.00	−0.37	0.869	0.705	5.4	0.878

Table 26: Fraction of the rescued cell answered correctly under each injected component. Synthetic control, multiple choice, Qwen3-1.7B, fp32, $n = 358$, rescued cell $n_R = 82$. Bootstrap 95% CIs over items are ± 0.09 to ± 0.11 at these magnitudes and are given in the appendix. $\bar{d}$ is the mean content direction over items, restored to the typical per-item norm; d_j is another item's content vector, drawn from the same or a different conversion family. The row that licenses the name "content component" is g.

injected at the last prompt position	L17–20	L21	L24	L26	L27
h_{skill} (replace outright; upper ref.)	0.29–0.45	0.988	0.963	1.000	1.000
h_{skill} into the wrong-document prompt	0.51–0.77	0.976	0.963	1.000	1.000
$h_{\text{no}} + 0.5\, d_i$	0.15–0.27	0.232	0.280	0.317	0.268
$h_{\text{no}} + d_i$ (content)	0.32–0.44	0.500	0.695	0.805	0.780
$h_{\text{no}} + 2\, d_i$	0.43–0.67	0.841	0.976	0.976	0.951
$h_{\text{no}} + g_i$ (presence)	0.00–0.04	0.098	0.098	0.098	0.098
$h_{\text{no}} + \bar{d}$ (one shared direction)	0.24–0.32	0.122	0.049	0.085	0.098
$h_{\text{no}} + d_j$ (same family)	0.29–0.37	0.268	0.244	0.317	0.293
$h_{\text{no}} + d_j$ (other family)	0.18–0.23	0.073	0.049	0.049	0.049

the model, although it does not establish that mechanism. A separate caveat is intervention size: skill spans in the screened subset contain 365–1,630 positions, far more than a single-position probe. §4.4 separates the number of written positions from their arrangement.

The pre-registered width-proportional prediction places Qwen3-0.6B's rank knee near 12. Its observed recovery at $k = 16, 32, 64, 128$ is 0.00, 0.12, 0.79, 0.67, giving $k^* = 43$ [39, 52], close to Qwen3-8B's 47 [43, 55] despite the fourfold width difference. On TheoremQA, the corresponding knees are 52 [46, 62] and 50 [39, 67], again inconsistent with proportional scaling. Their participation ratios are also similar in scale (65–75 versus 70–99 over layers 0–20, excluding unusually large activations). Mistral, with the same width as Qwen3-8B, gives $k^* = 72$ [46, 94] on MedCalc and 64 [50, 79] on TheoremQA: higher than Qwen3-8B on both tasks, but with overlapping intervals. Its curve is also less sharply defined, since recovery keeps rising past $k = 128$ (0.54; 0.72 at 256) and reaches its untruncated 0.84 at $k = 384$, against 256 for Qwen3-8B. Its bottom-k controls reach 0.11–0.21. The measured rank requirement is therefore not determined by hidden width alone, while a difference between families is not established.

How to read an arm's number. Interventions are reported both as raw accuracy and as the fraction of the range an intervention could possibly recover,

$$\rho(\text{arm}) = \frac{\text{acc(arm)} - \text{acc(receiver)}}{\text{acc(donor)} - \text{acc(receiver)}}, \tag{6}$$

where the receiver is the unpatched target prompt and the donor contains the source document. Normalisation matters when baselines differ: the four configurations in Figure 11 have donor accuracies of 0.550, 0.467, 1.000 and 0.950, with receiver accuracies from 0.000 to 0.350. Raw accuracies of 0.408 and 1.000 correspond to recovery fractions of 0.87 and 1.00 in their respective settings. Transfer recovery uses Eq. 3; knockout survival instead reports the fraction of the knockout run's rescued items that remain correct.